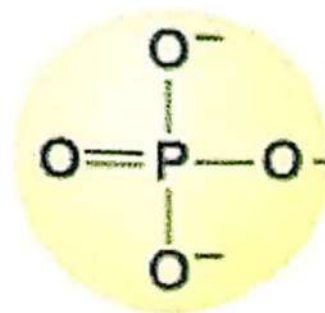
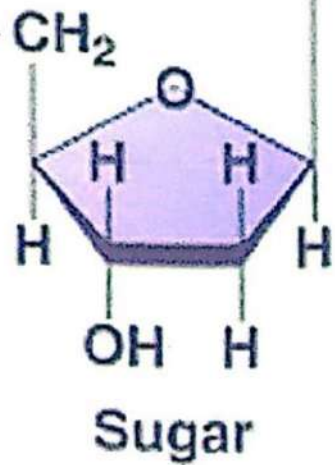


Hi Gene!





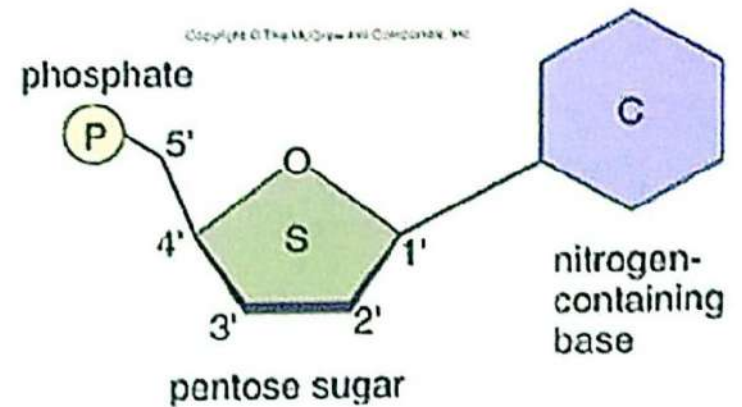
Phosphate group



Sugar



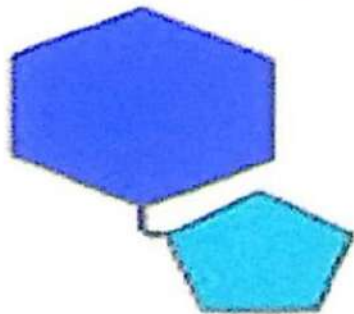
Nitrogenous base (adenine)



Nucleosides and Nucleotides

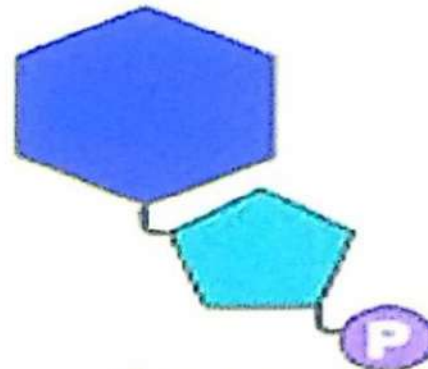
Nucleoside: If a molecule is composed of a purine or pyrimidine base and a ribose or deoxyribose sugar. If a **phosphate group** is added to the nucleoside, the molecule is now called a **nucleotide**.

Nucleoside



Base
+
Sugar

Nucleotide



Base
+
Sugar
+
Phosphate

Nucleotides

Ribonucleotide

Deoxyribonucleotide

N.B+Ribose sugar+P

N.B+Deoxyribose sugar+P

A Adenlic acid

Deoxyadenlic acid

C Cytidylic acid

Deoxcytidylic acid

G Guanylic acid

Deoxguanylic acid

U Uridylic acid

Deoxythymidylic acid

T

Nucleotides

Ribonucleotide

N.B+Ribose suger+P

A Adenlic acid

C Cytidylic acid

G Guanylic acid

U Uridylic acid

T

Deoxyribonucleotide

N.B+Deoxyribose suger+P

Deoxyadenlic acid

Deoxcytidylic acid

Deoxguanylic acid

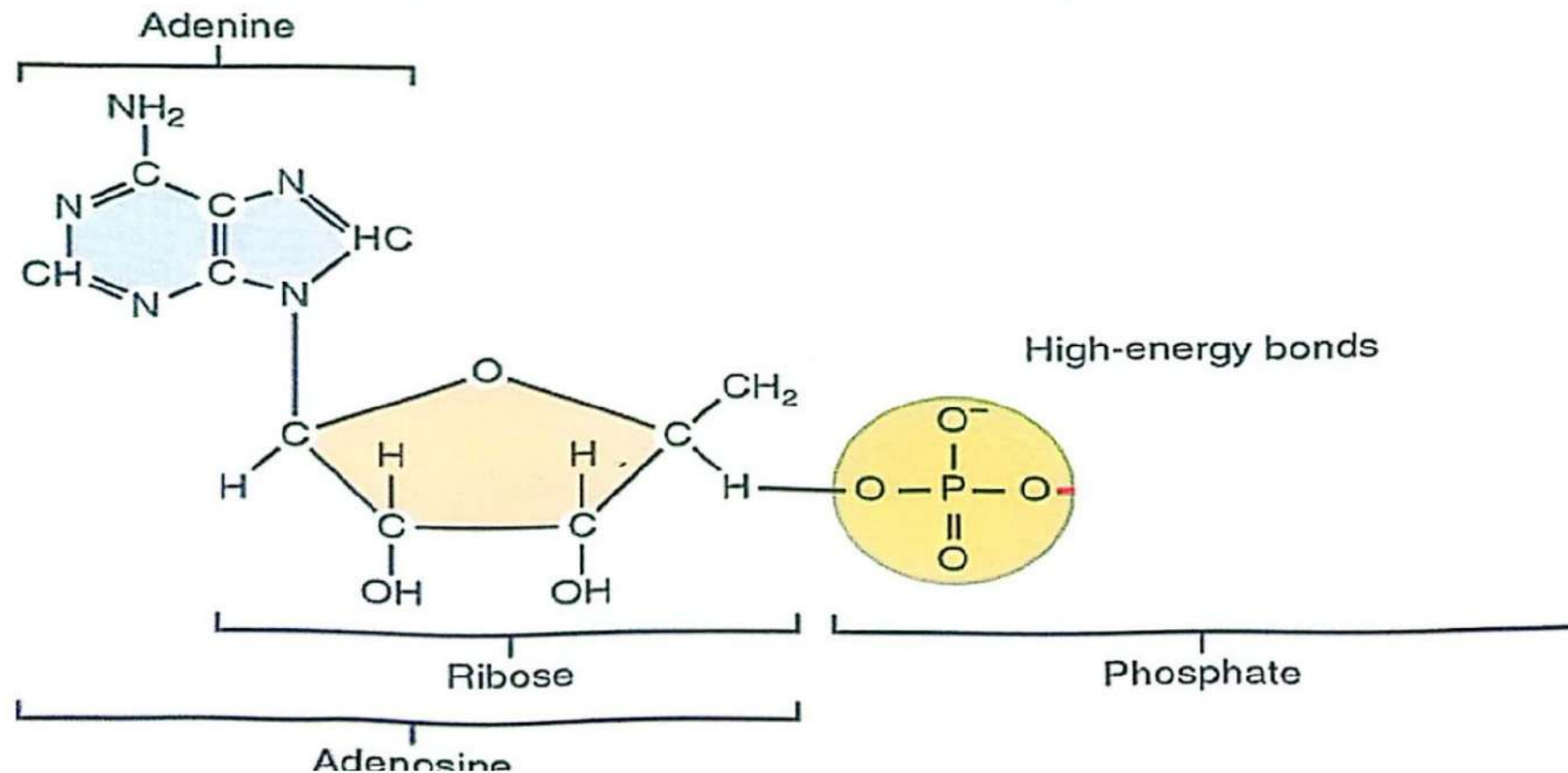
Deoxythymidylic acid

Nucleoside Diphosphates and Triphosphates:

Mononucleotide = Nucleotide = Nucleoside + P

Mononucleotides = nucleosides monophosphates.

The addition of one or two phosphate groups results in nucleoside diphosphate and nucleoside triphosphate

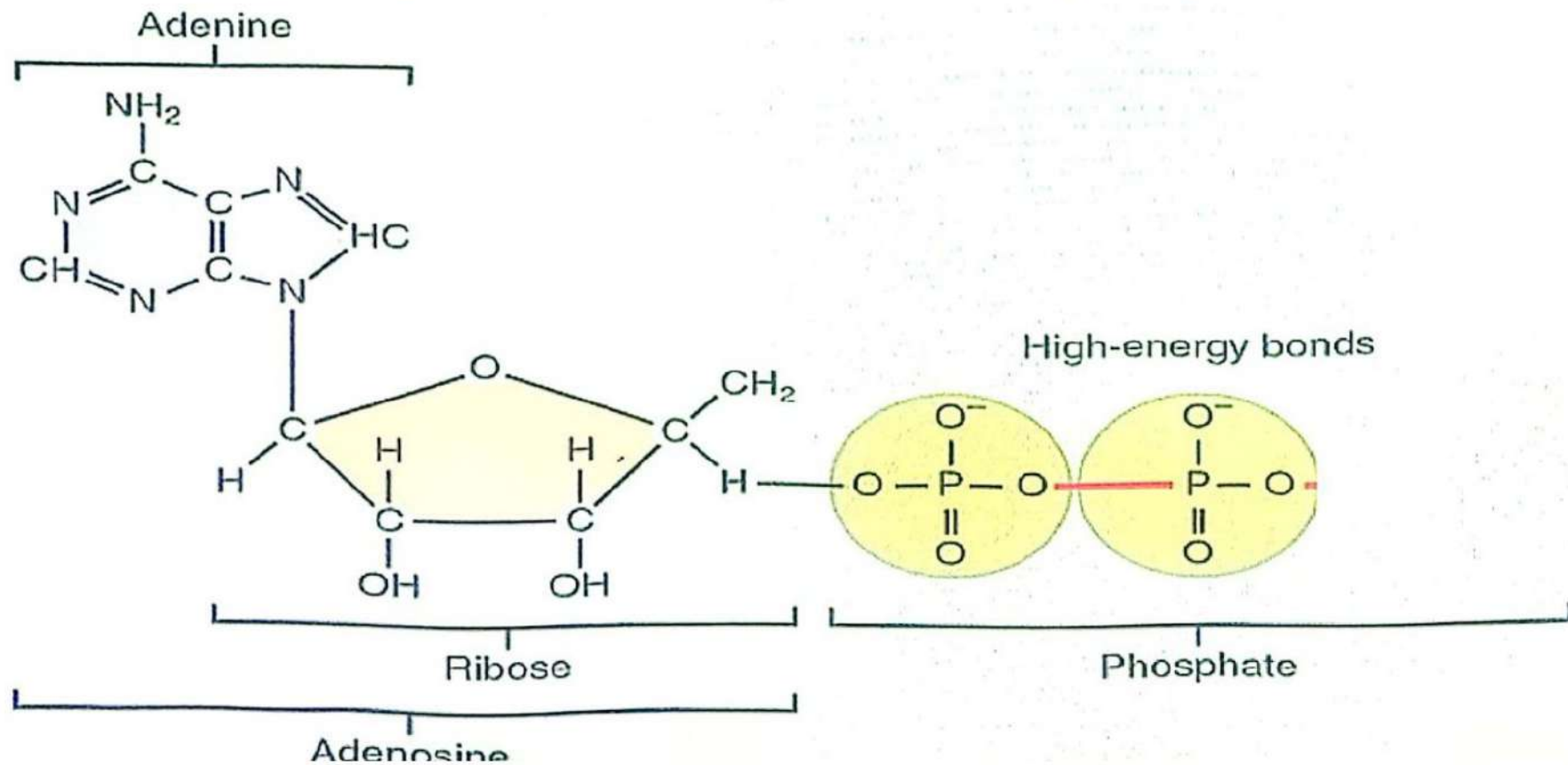


Nucleoside Diphosphates and Triphosphates:

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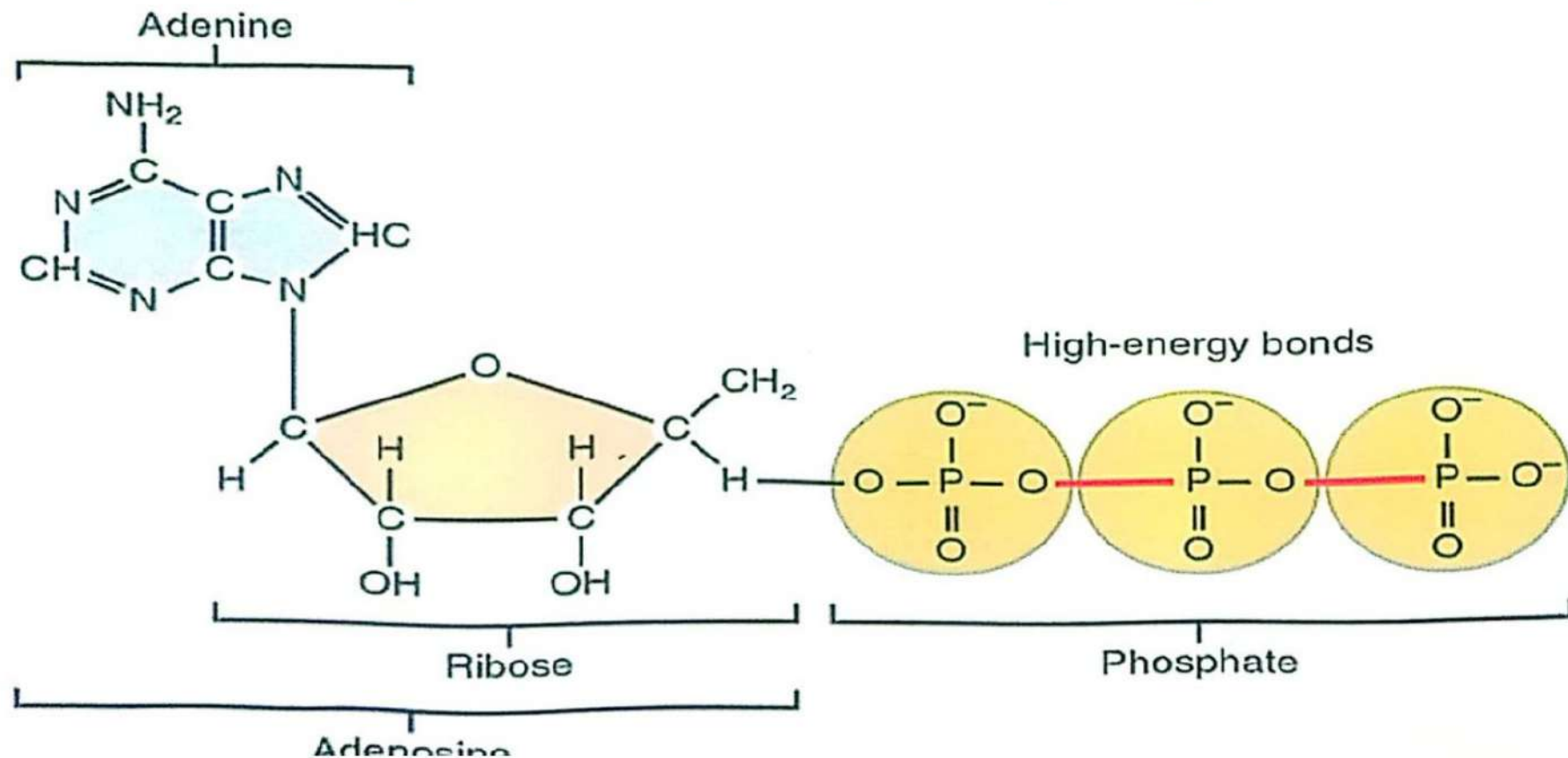


Nucleoside Diphosphates and Triphosphates:

Mononucleotide = Nucleotide = Nucleoside + P

Mononucleotides = nucleosides monophosphates.

The addition of one or two phosphate groups results in nucleoside diphosphate and nucleoside triphosphate



Nucleoside Diphosphates and Triphosphates:

Importance of Nucleoside triphosphate

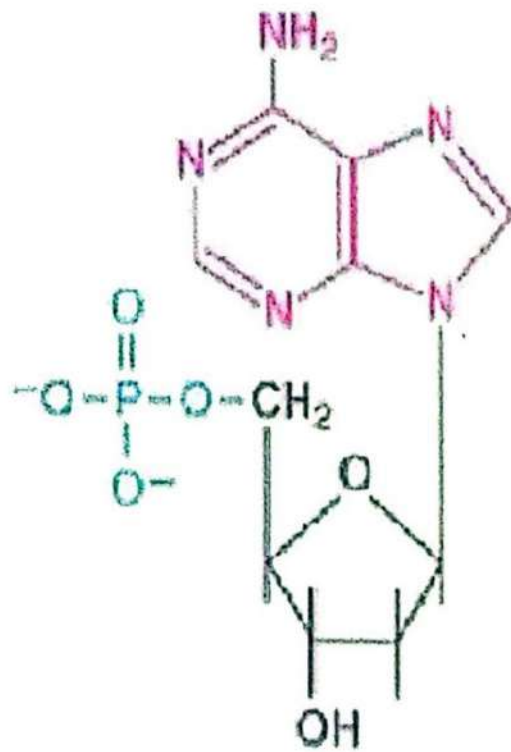
1- It serves as the precursor molecule during nucleic acid synthesis within the cell (ATP, GTP, CTP, TTP).

2- The triphosphates:

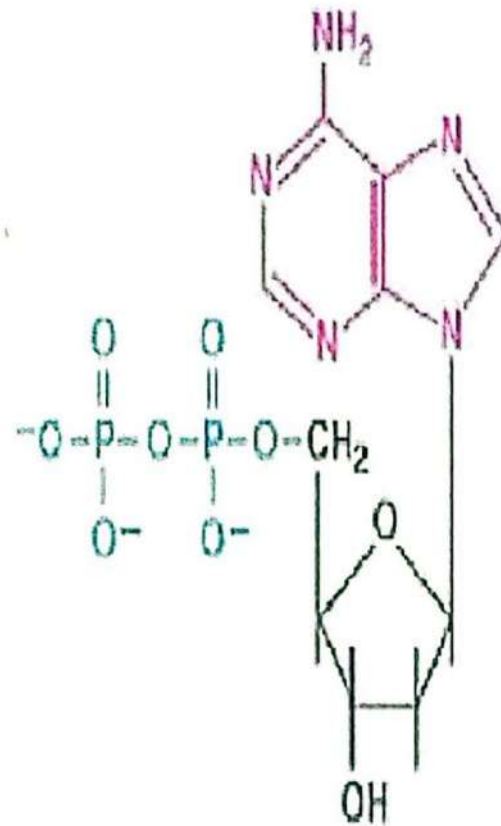
adenosine triphosphate (ATP) and
guanosine triphosphate (GTP)

are involved in the bioenergetics of the cell,
because of the large amount of energy used or released by the
addition or removal of the terminal phosphate group

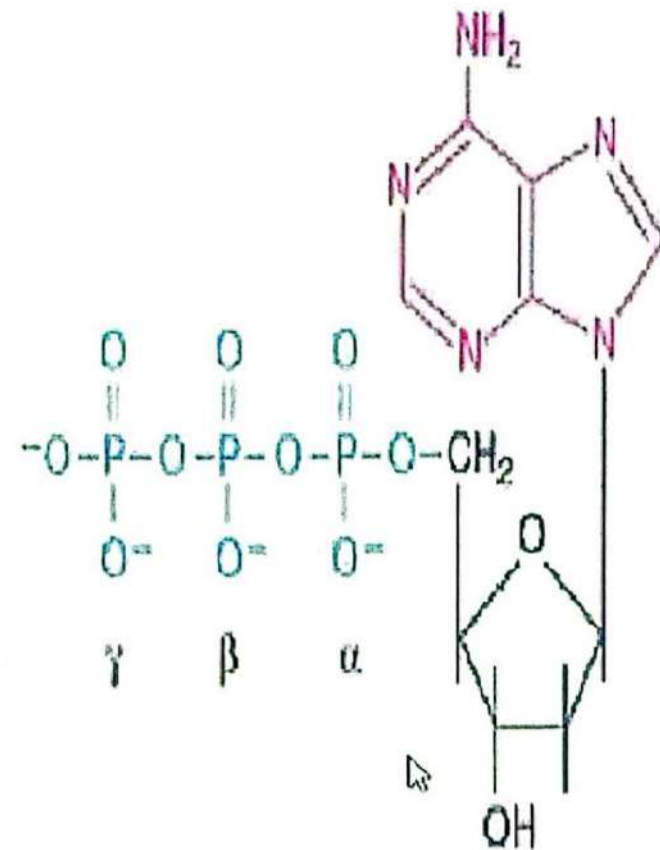




Deoxyadenosine-5'-
monophosphate (dAMP)



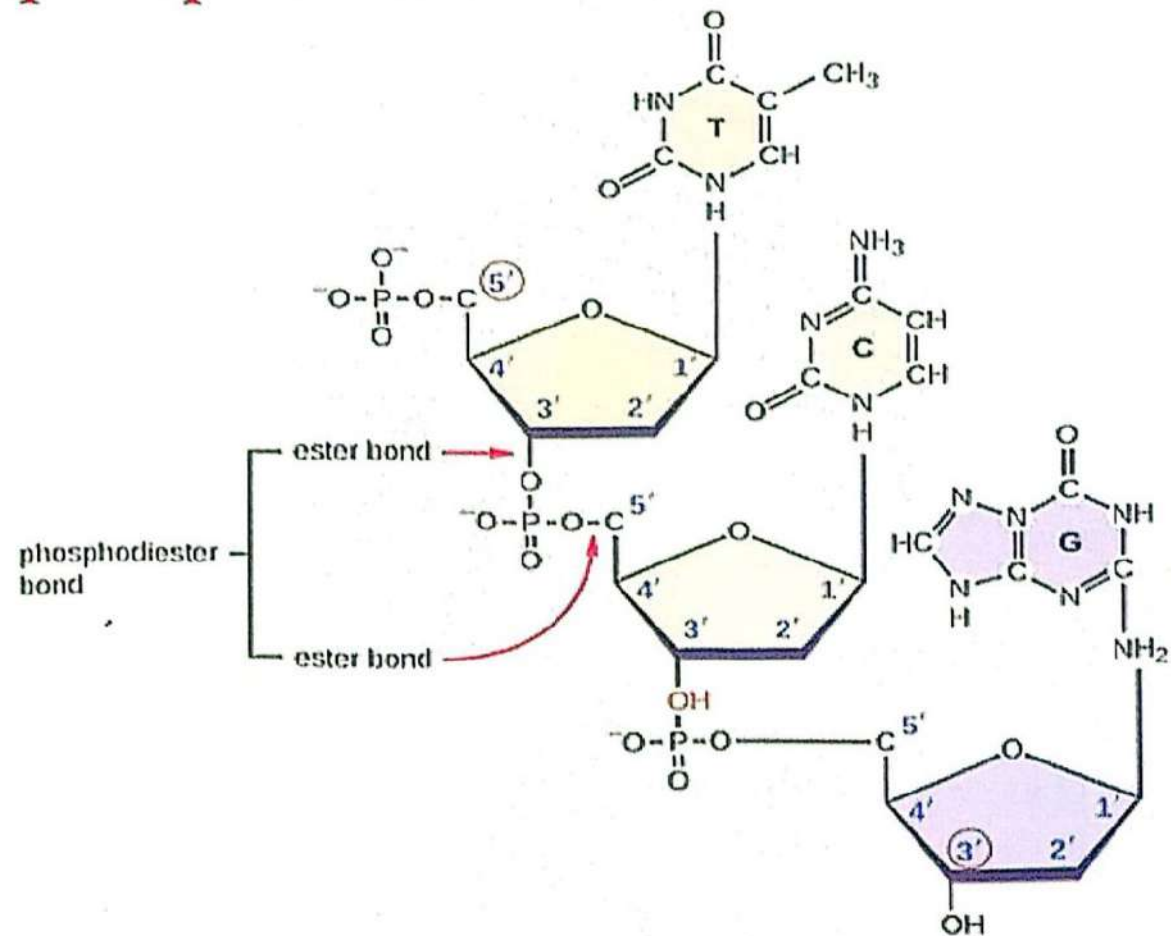
Deoxyadenosine-5'-
diphosphate (dADP)



Deoxyadenosine-5'-
triphosphate (dATP)

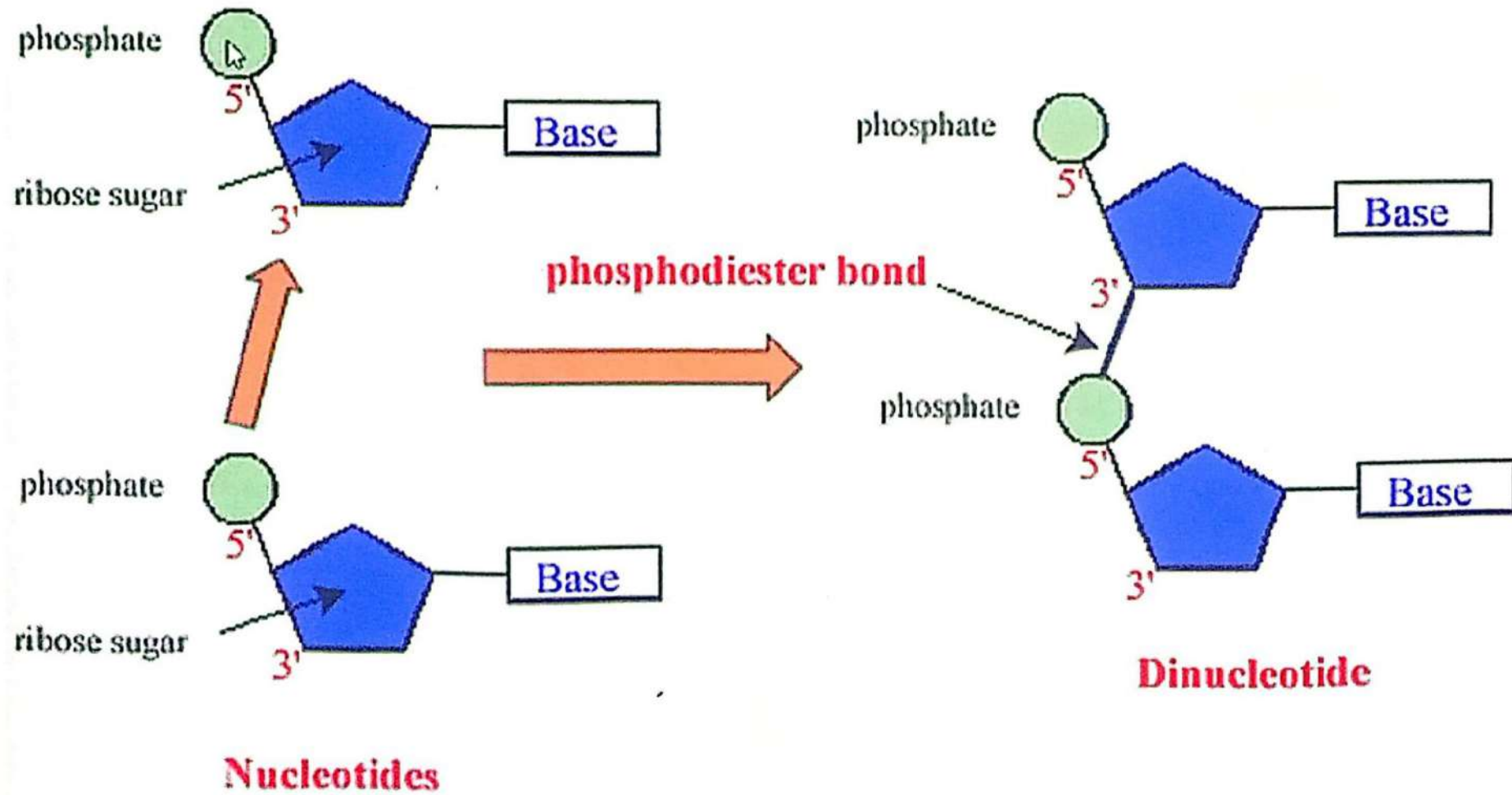
Polynucleotides

The linkage between the two mononucleotides requires the linkage of **a phosphate** group to the two sugar molecules by **phosphodiester bond**



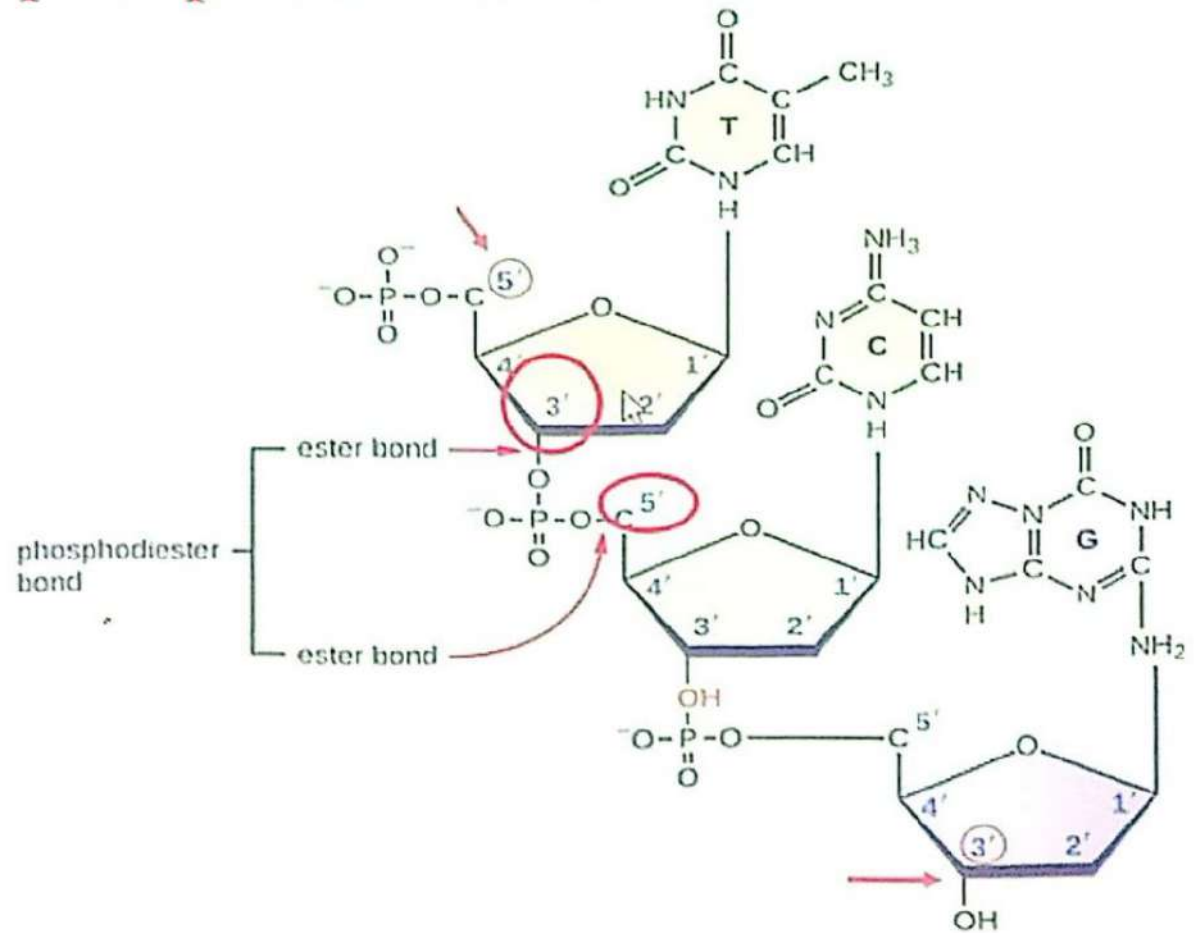
Phosphodiester bond

Polynucleotide formation



Polynucleotides

The linkage between the two mononucleotides requires the linkage of a phosphate group to the two sugar molecules by phosphodiester bond



Phosphodiester bond

Polynucleotides

Phosphoric acid has been joined to two alcohols (the OH groups on the two sugars) by an ester linkage on both sides.

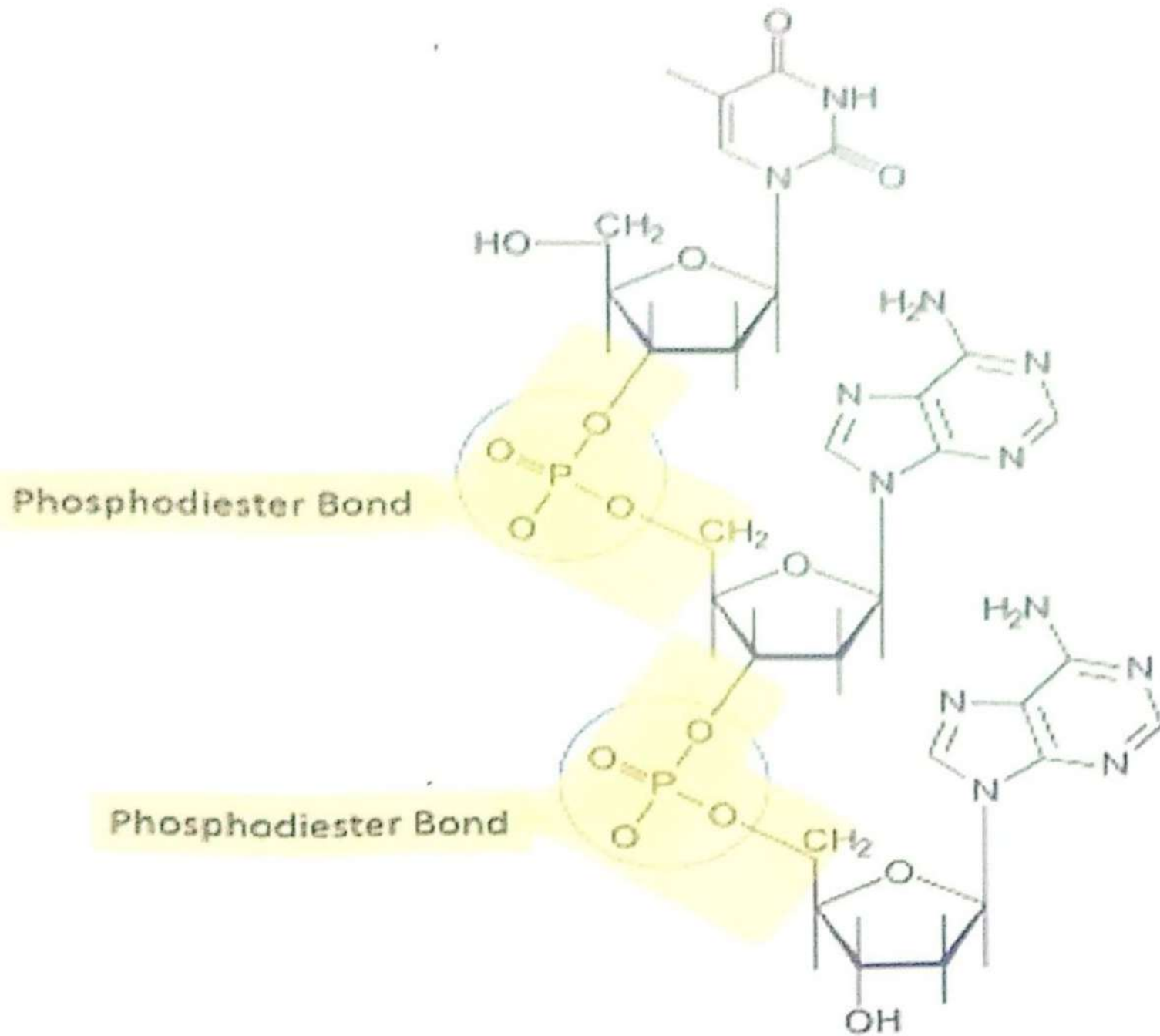
phosphodiester bonds in DNA (or RNA).

Each structure has a C-5' end and a C-3' end.

The joining of a long chain of nucleotides, is called polynucleotide

Long polynucleotide chains would account for the most important property of DNA.

Genetic variation



The Structure of DNA

In 1953, Watson and Crick proposed that the structure of DNA is in the form of a ***double helix***.

The available data which preceded and led to the development of this proposal came mainly from two sources:

1. Base composition and
2. X-ray crystallography studies of DNA.
3. Watson and Crick succeeded by the use of model building which confirmed the available data.

The Structure of DNA

1. Base Composition of DNA

Chargaff and his associates used quantitative methods to determine

the amounts of the four nitrogenous bases

after their separation from DNA samples obtained from different sources as

Human, thymus, Human, liver Human, sperm

The Structure of DNA

Source	Base composition								$\frac{A+T}{G+C}$
	A	T	G	C	A/T	G/C	A+G	T+C	
Human, thymus	30.9	29.4	19.9	19.8	1.05	1.01	50.8	49.2	1.52
Human, liver	30.3	30.3	19.5	19.9	1.00	0.98	49.8	50.2	1.53
Human, sperm	30.7	31.2	19.3	18.8	0.98	1.03	50.0	50.0	1.62
Herring, sperm	27.8	27.5	22.2	22.6	1.01	0.98	50.0	50.1	1.23
Rat, bone marrow	28.6	28.4	21.4	21.5	1.01	1.00	50.0	49.9	1.30
E. coli	24.7	23.6	26.0	25.7	1.04	1.01	50.7	49.3	0.93
Phage T ₇	26.0	26.0	24.0	24.0	1.00	1.00	50.0	50.0	1.08

$$A = T$$

$$G = C$$

$$\text{Sum of Purines} = \text{Sum of Pyrimidines}$$

$$A+G = T+C$$

$$\frac{A+G}{T+C} = 1$$

$$\frac{A+T}{G+C} \neq 1$$

3. The ratio of $(A+T)/(C+G)$ does not necessarily one, and **varies greatly** among species
4. The base composition of DNA in any given tissue is **the same** (within experimental error) for the same species

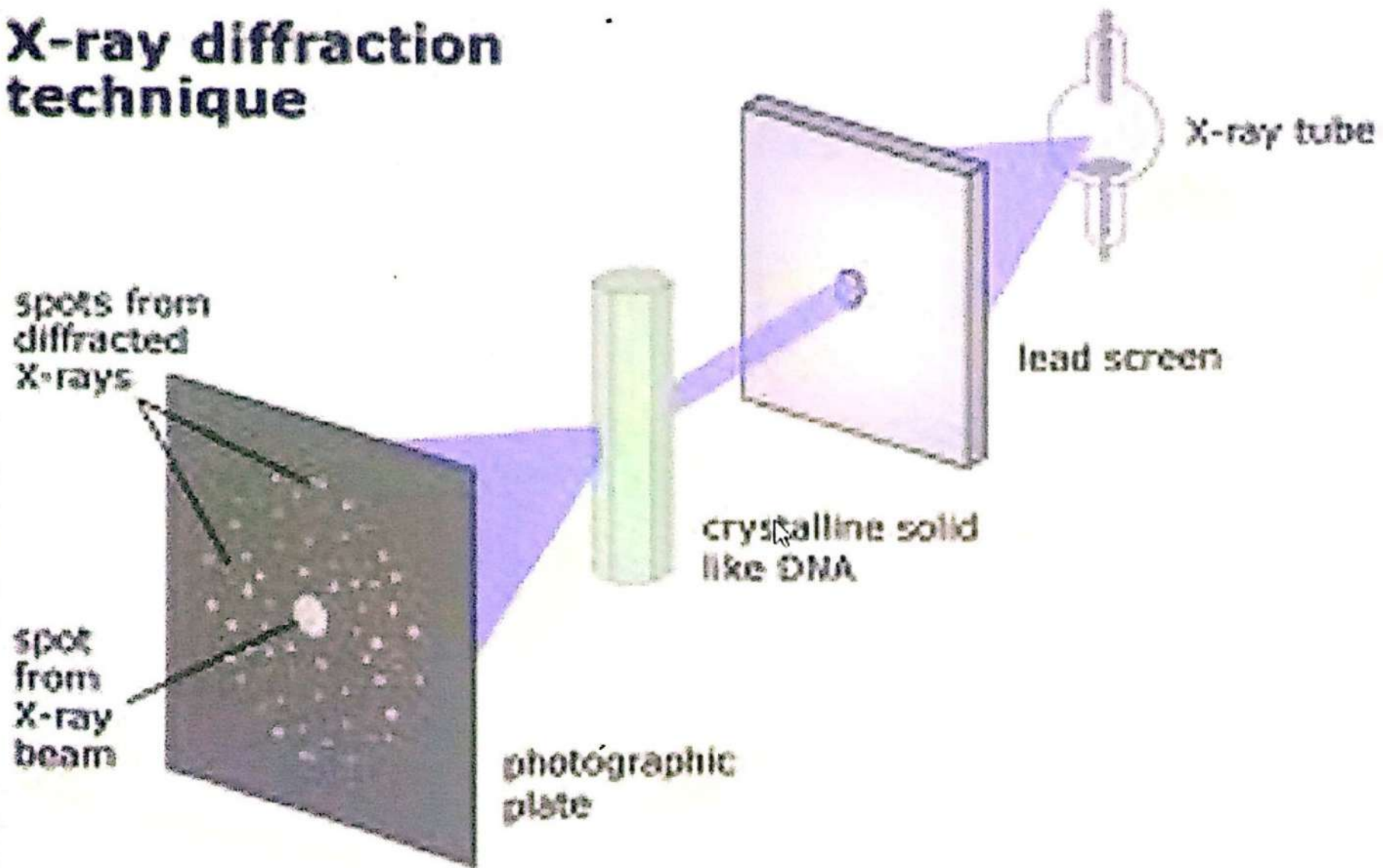
X-ray Crystallography of DNA

X-ray crystallography of highly purified DNA samples were obtained by both Pauling and Franklin.

They both proposed a helical structure.

The later author demonstrated two distinctive regularities (or periodicities) of **0.34 nm** (3.4 Å) and **3.4 nm** (34 Å) along the axis of DNA

X-ray diffraction technique



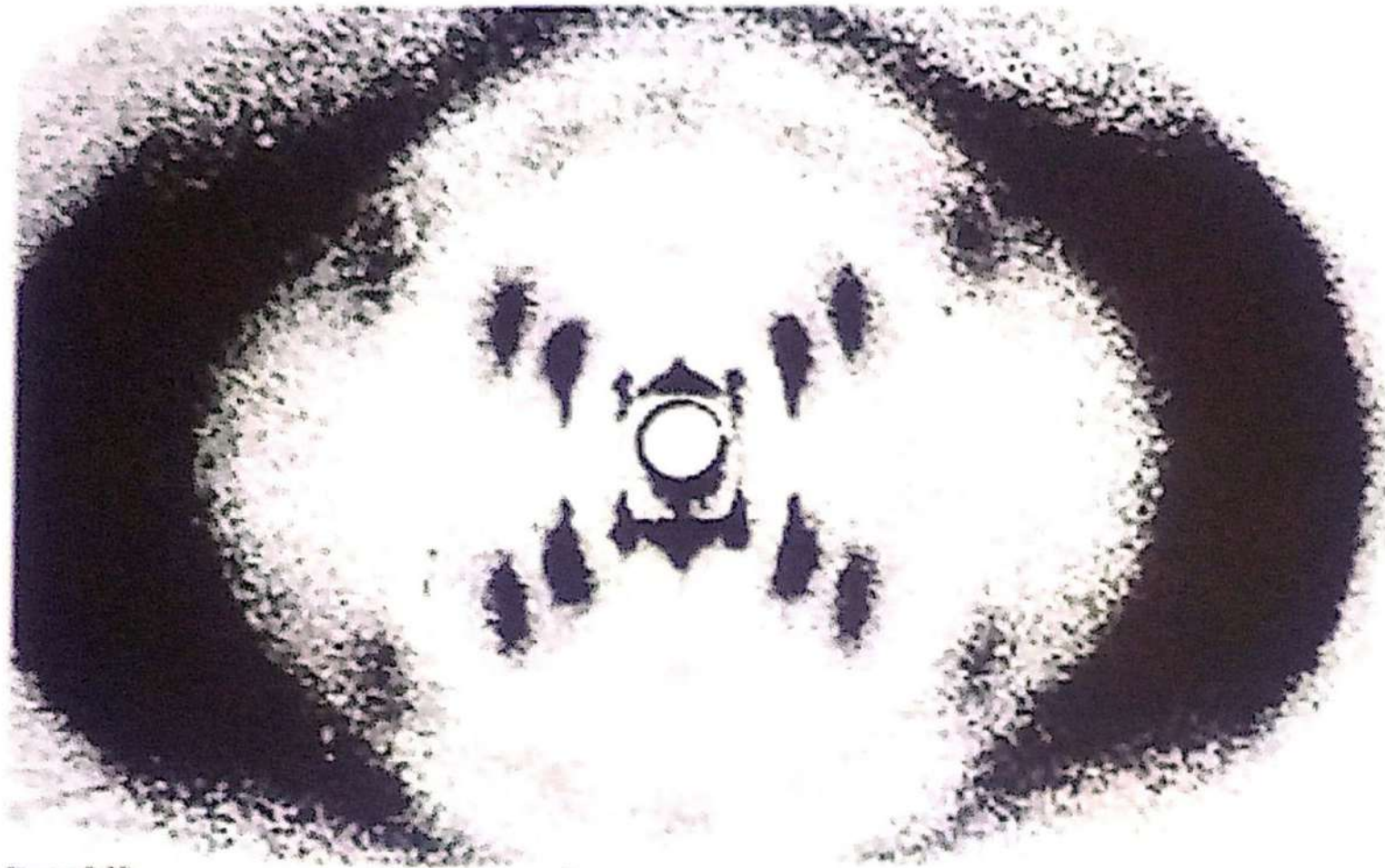
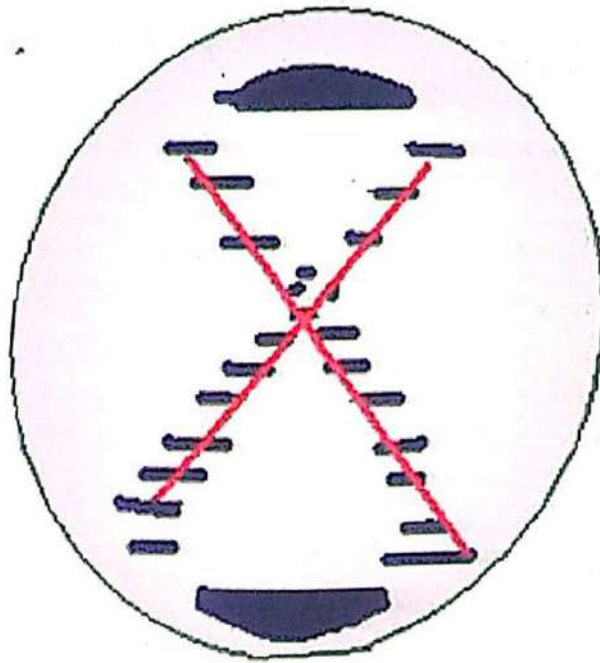


Figure 6-12

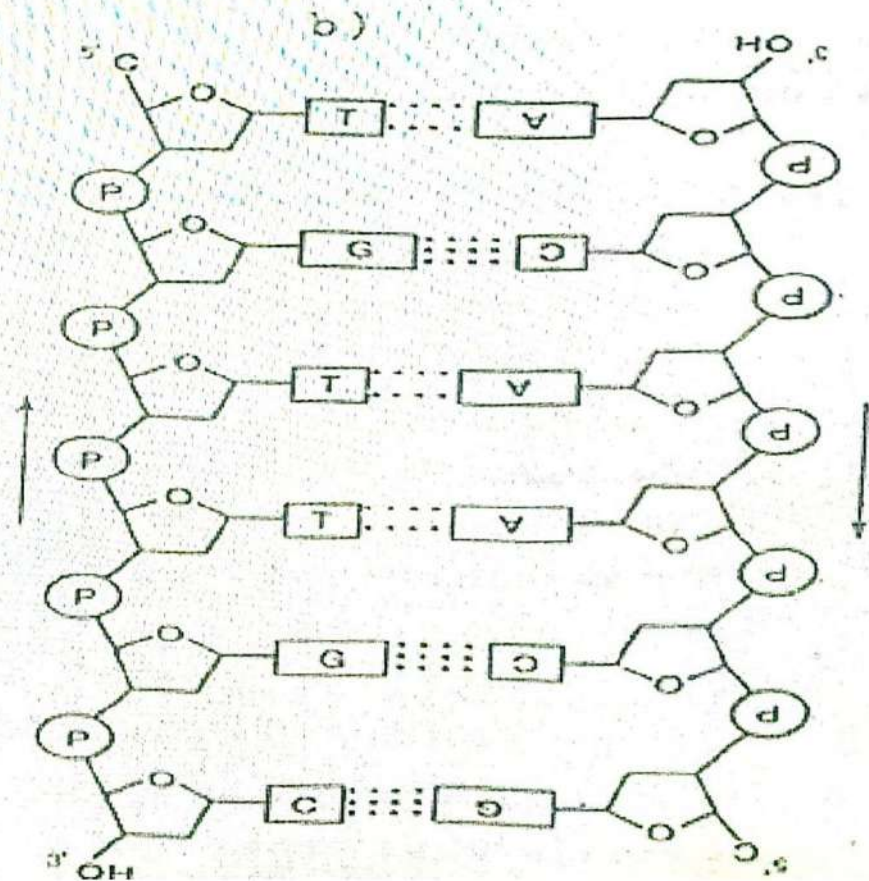
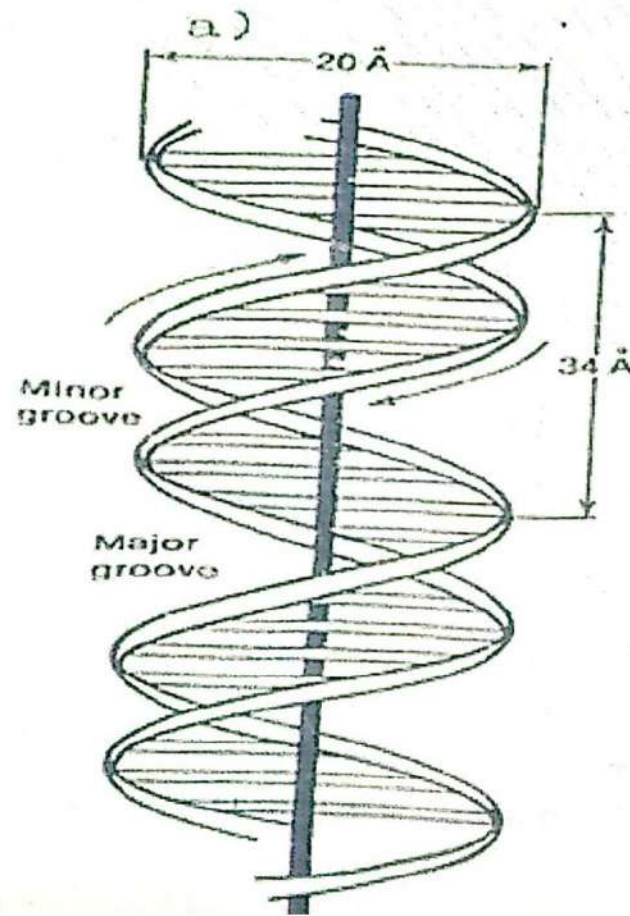
DeBorja's Photographs of the Human Body, 1968 Edition
© 1968 by R. DeBorja, used by permission

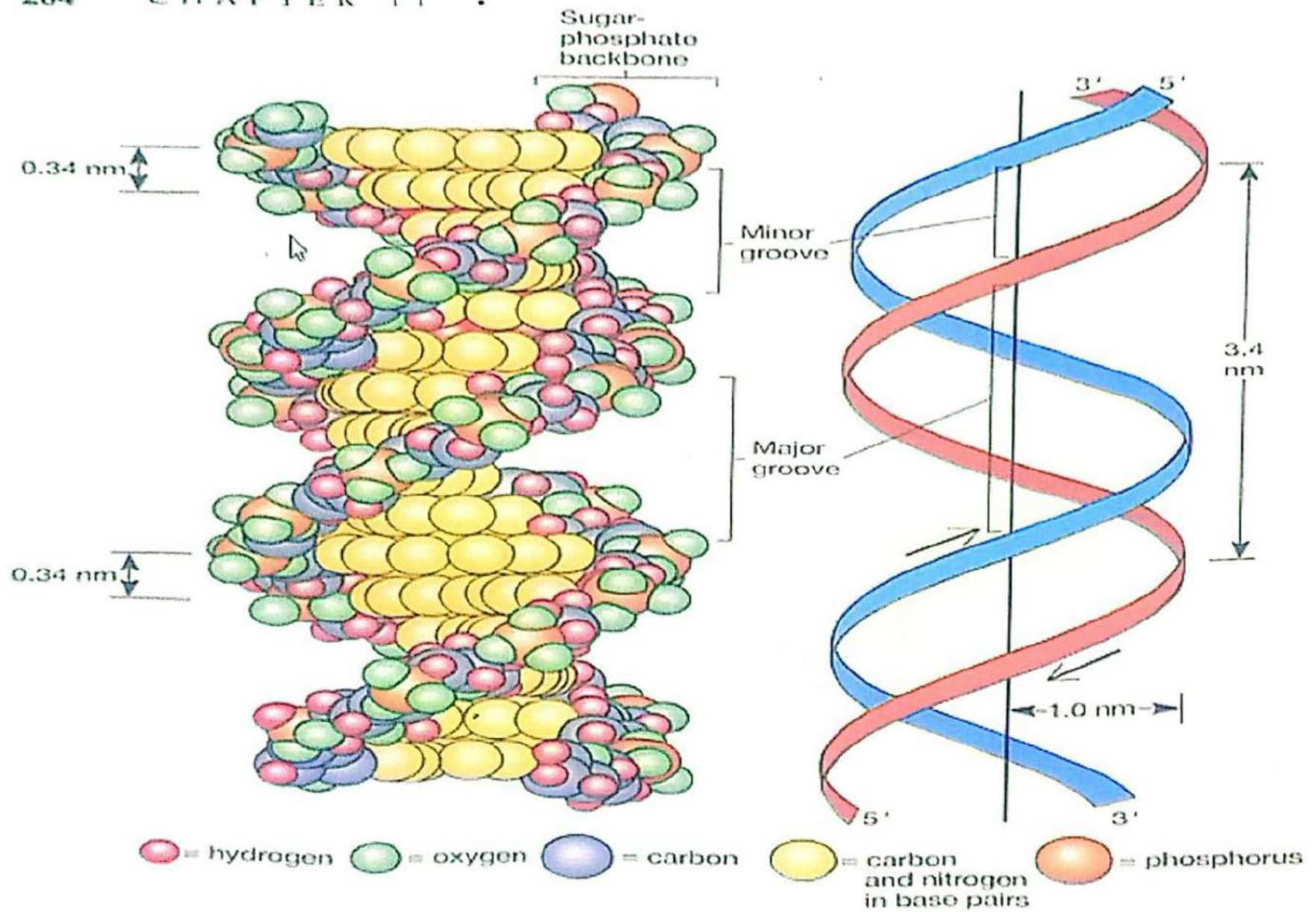


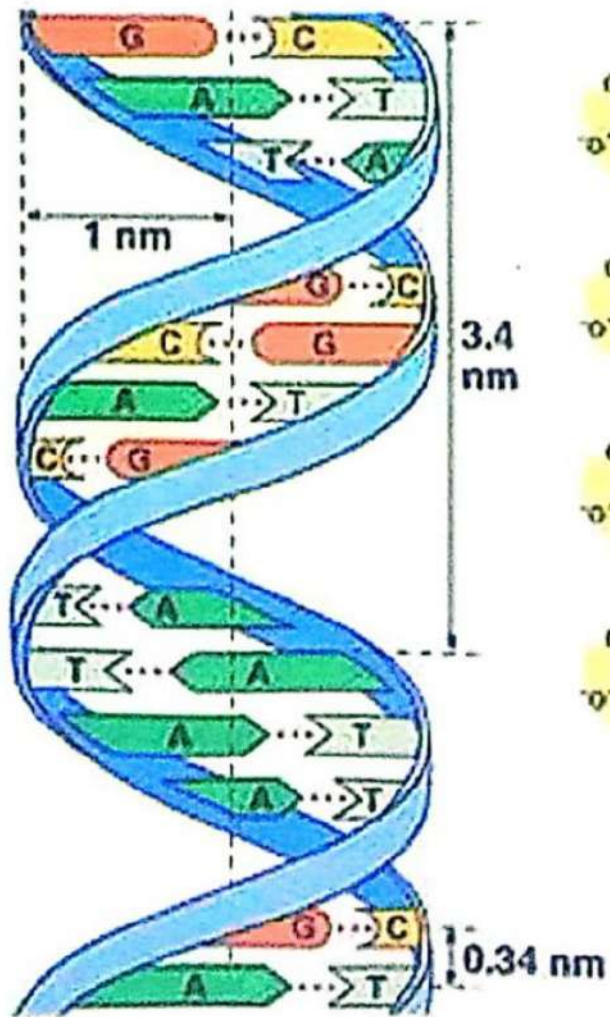
diameter is same

The Watson-Crick Model

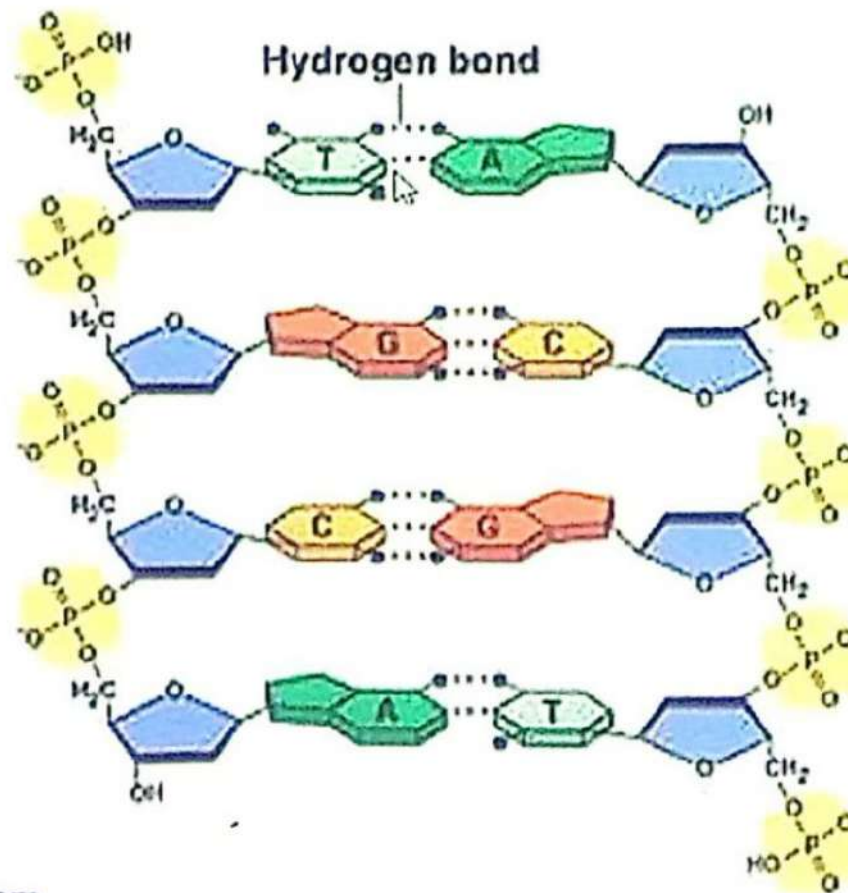
By building models to conform with the information supplied by the base composition data and X-ray crystallography, Watson and Crick arrived at the double-helical model of DNA



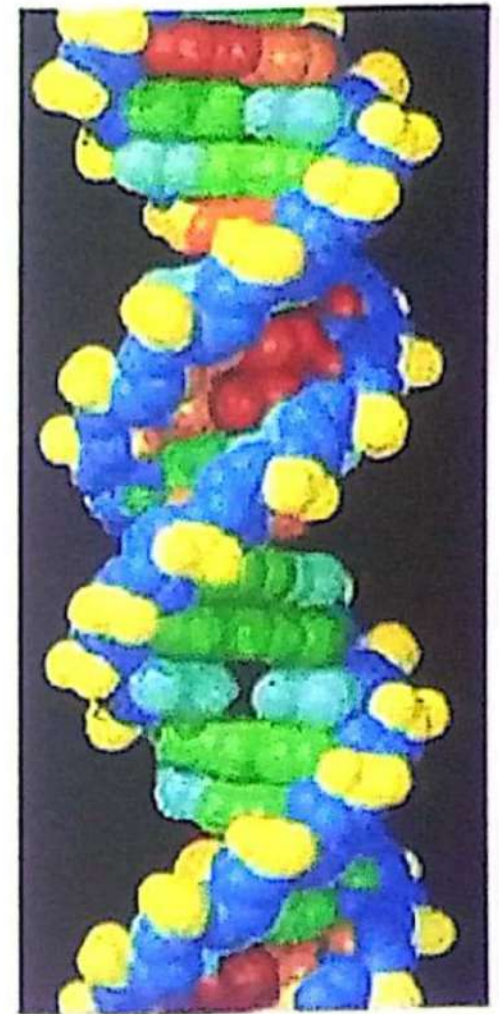




(a)



(b)



(c)

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The Watson-Crick Model

This model has the following characteristics

1. **Two right-handed helical** polynucleotide chains are coiled around a central axis
2. The two chains are **antiparallel**; i.e., one is upside down with respect to the other; their C-5_ to C-3_ orientations are in opposite directions .
3. The base of both chains are stacked on one another, **0.34 nm apart**.
4. Each complete turn of the helix is **3.4 nm** long; thus, **10 bases** exist in each chain per turn

The Watson-Crick Model

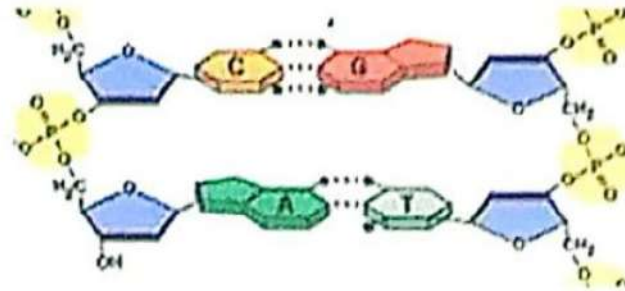
5. The nitrogenous bases of opposite chains are electrostatically attracted to one another as the result of the formation of *hydrogen bonds* specifically only A-T and C-G pairs are allowed.
6. In any segment of the molecule, alternating larger *major grooves* and smaller *minor grooves* appear along the axis
7. The double helix measures *2 nm (20 Å)* In diameter

One of the most significant features of the structure proposed by Watson and Crick is

1. The specificity of base pairing.

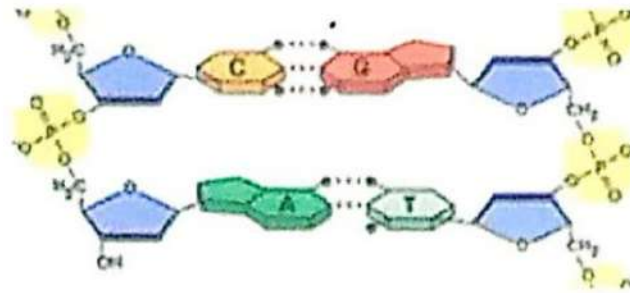
$A = T$ $G = C$.

a. when placed opposite each other, provided chemical stability necessary to hold the chains together through the hydrogen



3. The specific **A-T** and **G-C** base pairing is the basis for the concept of complementarity.

This means the chemical affinity provided by the hydrogen bonds between the bases.



3. The specific **A-T** and **G-C** base pairing is the basis for the concept of complementarity.

This means the chemical affinity provided by the hydrogen bonds between the bases.

Is very important for DNA replication and gene expression.

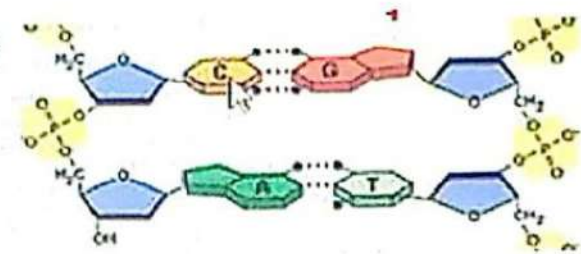
Stabilization of the helix

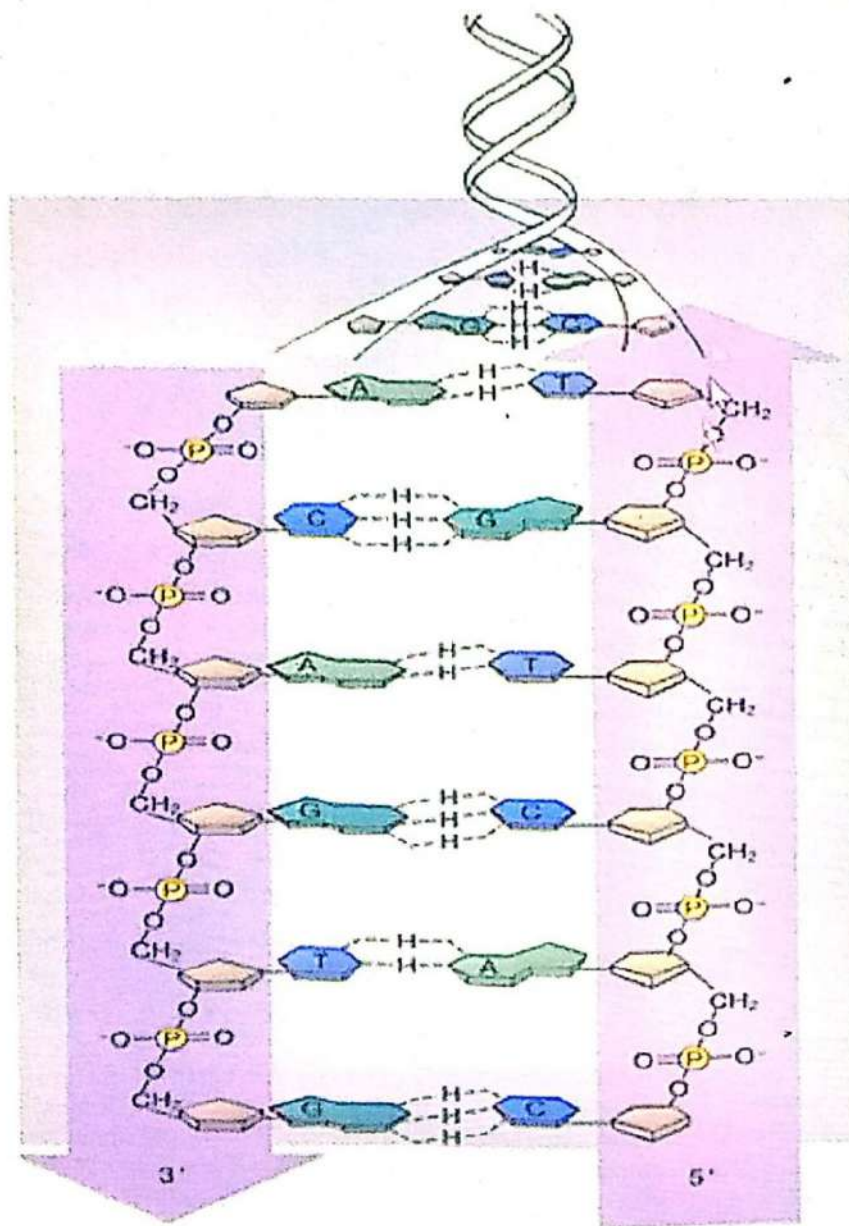
1. The *hydrogen bond* is a very weak electrostatic attraction.

A forms two hydrogen bonds with T

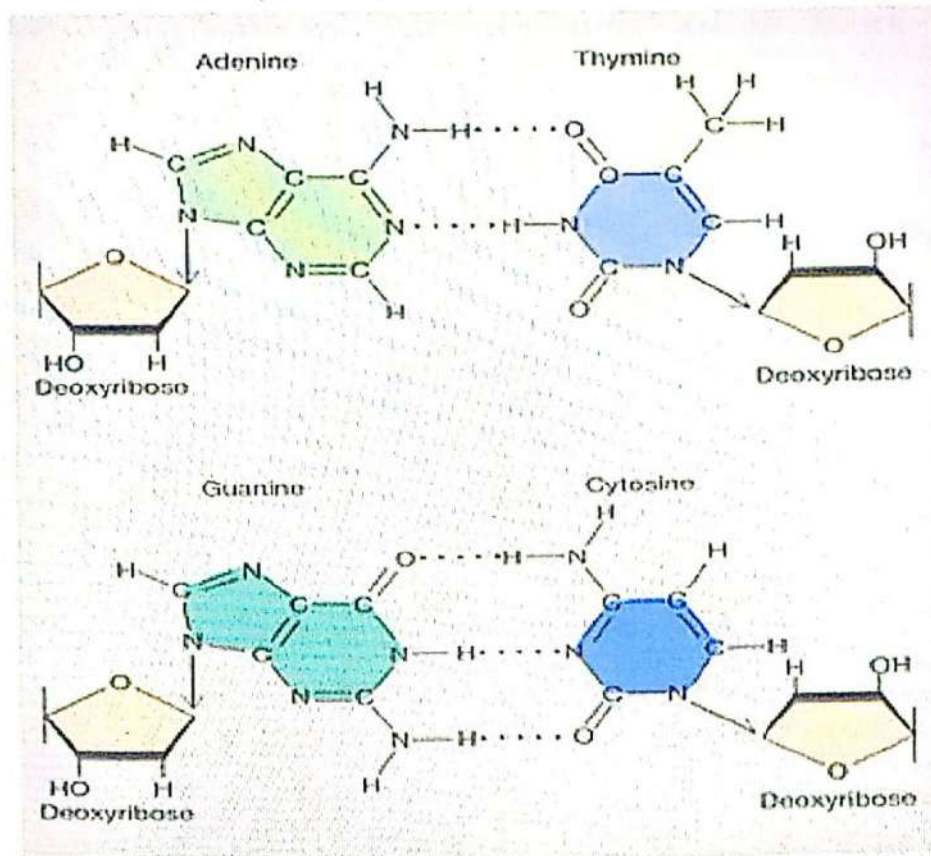
G forms three hydrogen bonds with C

two or three hydrogen bonds taken alone :
two or three thousands together can
provide great stability to the helix.





(a)



(b)

Figure 11-10 (a) Physical structure of DNA. The two sugar-phosphate chains run in opposite directions. This orientation permits the complementary bases to pair. (b) Diagram of the hydrogen bonding between base pairs: adenine (A) and thymine (T) (*top*) and guanine (G) and cytosine (C) (*bottom*). The AT pair has two hydrogen bonds; the GC pair has three.

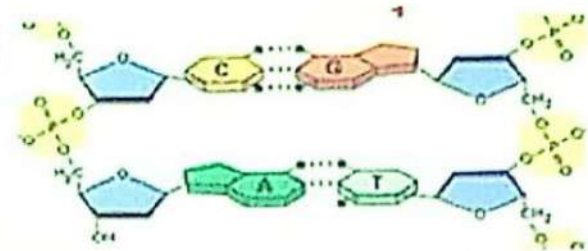
Stabilization of the helix

1. The *hydrogen bond* is a very weak electrostatic attraction.

A forms two hydrogen bonds with T

G forms three hydrogen bonds with C

two or three hydrogen bonds taken alone :
two or three thousands together can
provide great stability to the helix.



2. The arrangement of sugars and bases along the axis.

In the Watson-Crick model, the *hydrophobic* (water fearing) bases are stacked on the inside of the helix, thus protected from water.

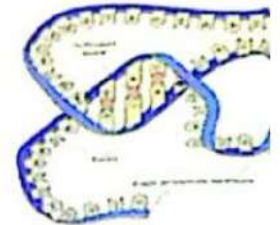
the *hydrophilic* (water loving) sugar phosphate backbone is on the outside

The Structure of RNA

Its structure is similar to DNA with the following exceptions

1. Ribose sugar in RNA replaces deoxyribose in DNA
2. The nitrogenous base **Uracil** in RNA replaces **Thymine** in DNA
3. Most RNA is a single-stranded polynucleotide chain

☐ RNA molecules sometimes fold back on themselves following their synthesis due to regions of complementarity leading to base pairing



☐ Some animal viruses which have RNA as their genetic material contain it in the form of double-stranded helix.

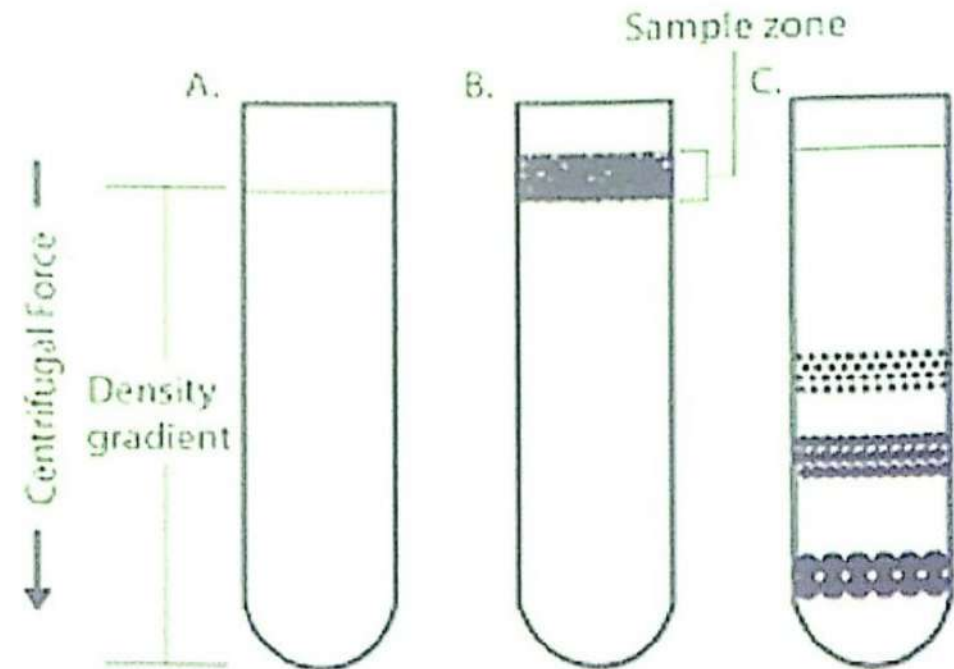
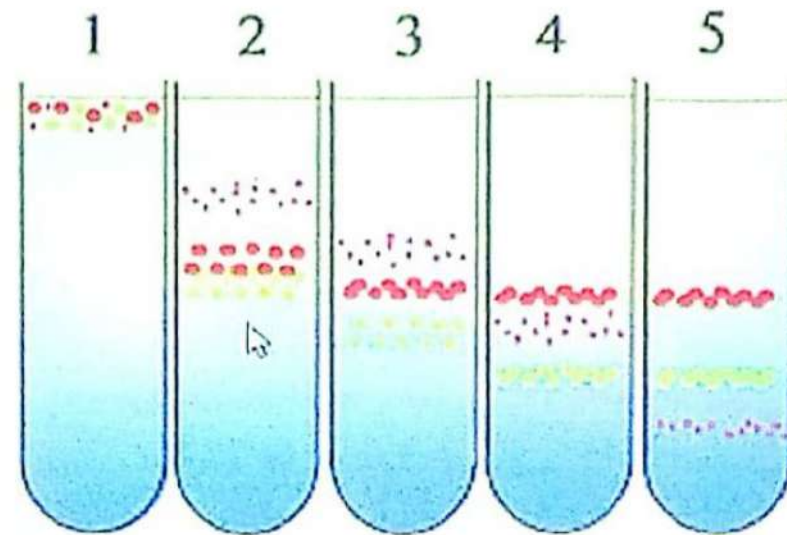
Classes of RNA

Each class of RNA is characterized by its

1. Size,
2. Genetic function.
3. Sedimentation behaviour in a centrifugal field,

on

of
tein
ular



Classes of RNA

Sedimentation behavior is measured by sedimentation velocity by ultracentrifugation in a sucrose gradient solution.

Sedimentation velocity depends on

Molecular density,

Mass

Shape of the molecule

its measure is called Svedberg coefficient (S).

Classes of RNA

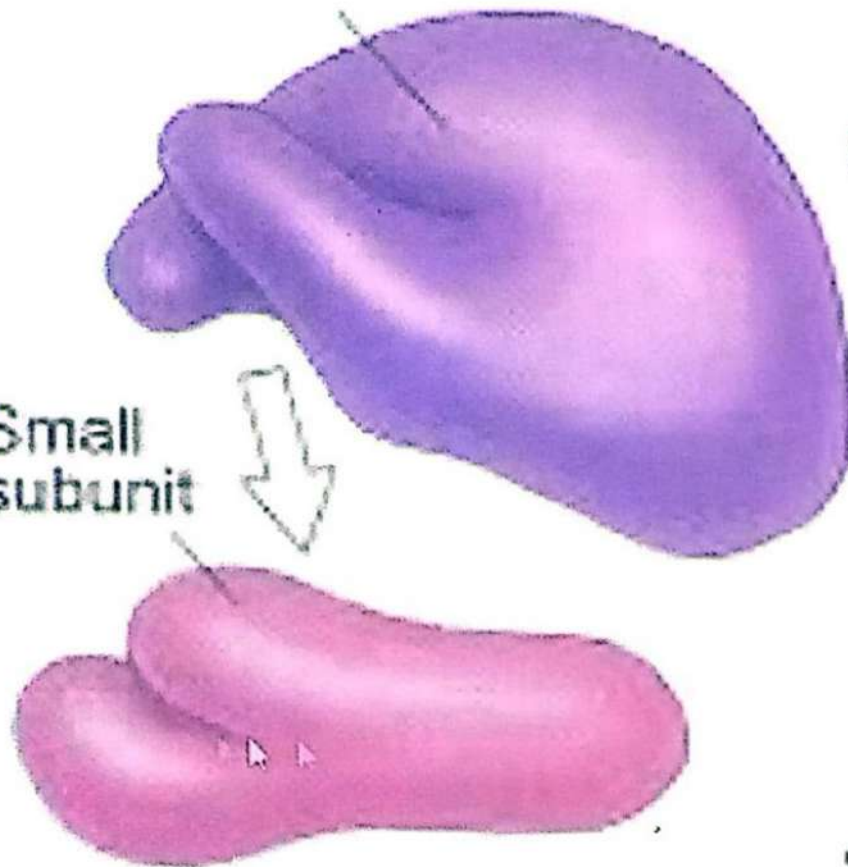
RNA type	Sedimentation Coefficient (S)	Molecular weight. (Dalton)	No. of nucleotides
Ribosomal RNA	5S (E. coli + mammals)	36,000	120
	16S (E. coli)	550,000-600,000	1700
	23S (E. coli)	1,100,000	3700
	18S (mammals)	700,000	1800
	28S (mammals)	1,800,000	5000
	5.8S (mammals)	50,000	
Transfer RNA	4S	23,000-30,000	75-90
Messenger RNA	6S- 25S	25,000-1,000,000	75-3000

Ribosomal RNA (rRNA)

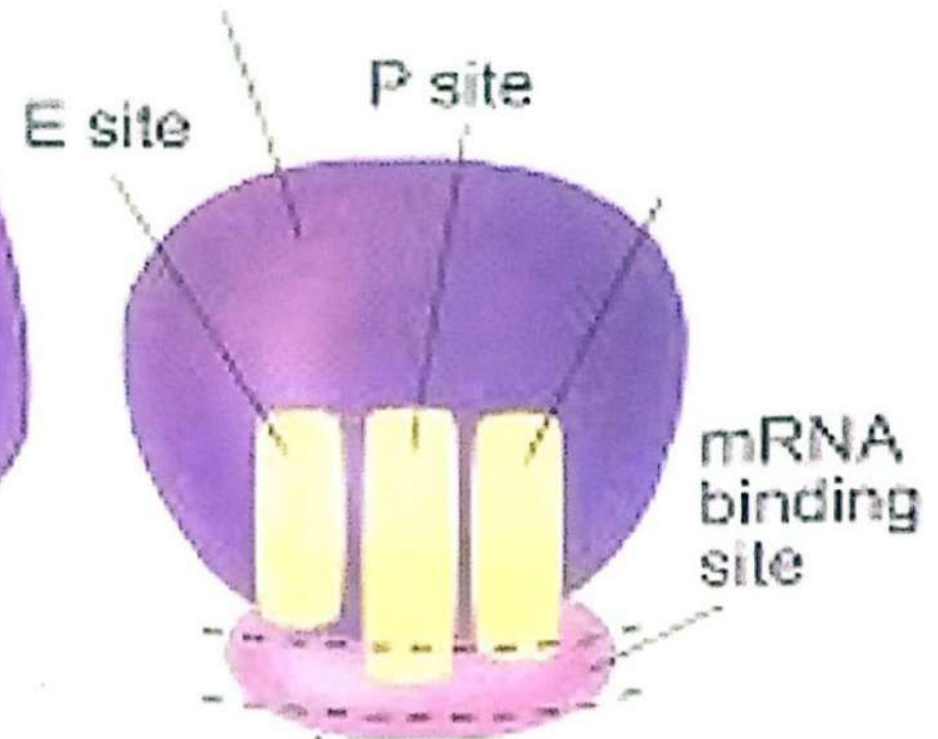
- ❖ The **largest** RNA molecule. About **80%** of all RNA in the cell.
- ❖ The various forms of rRNA found in prokaryotes and eukaryotes differ greatly in size.
- ❖ It is an important chemical component of ribosomes, which function in the synthesis of proteins, a process called **translation**.
- ❖ The large size of rRNA is related to its chemical interactions with proteins, forming the functional ribosomes,

Large subunit

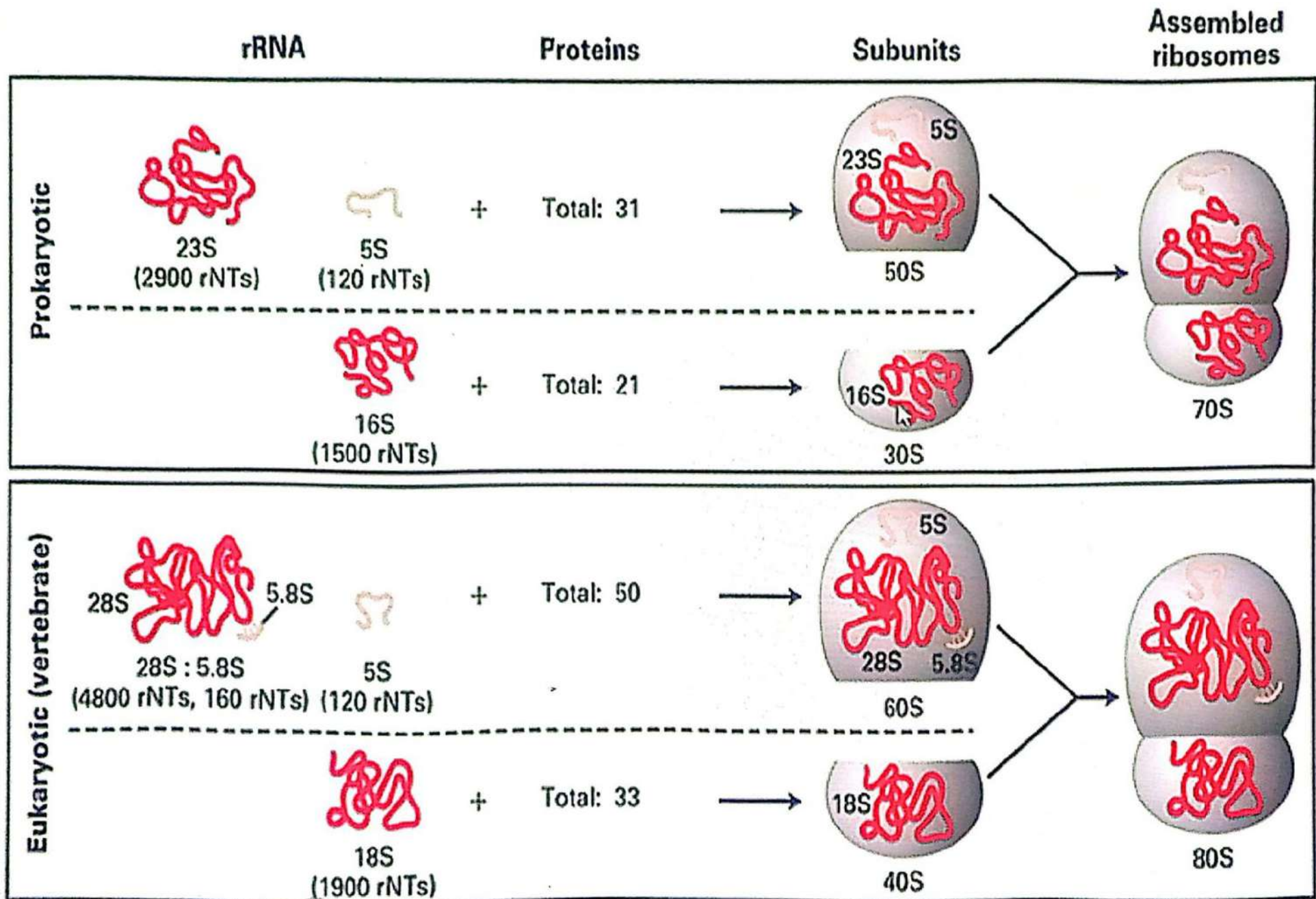
Small subunit



Large ribosomal subunit

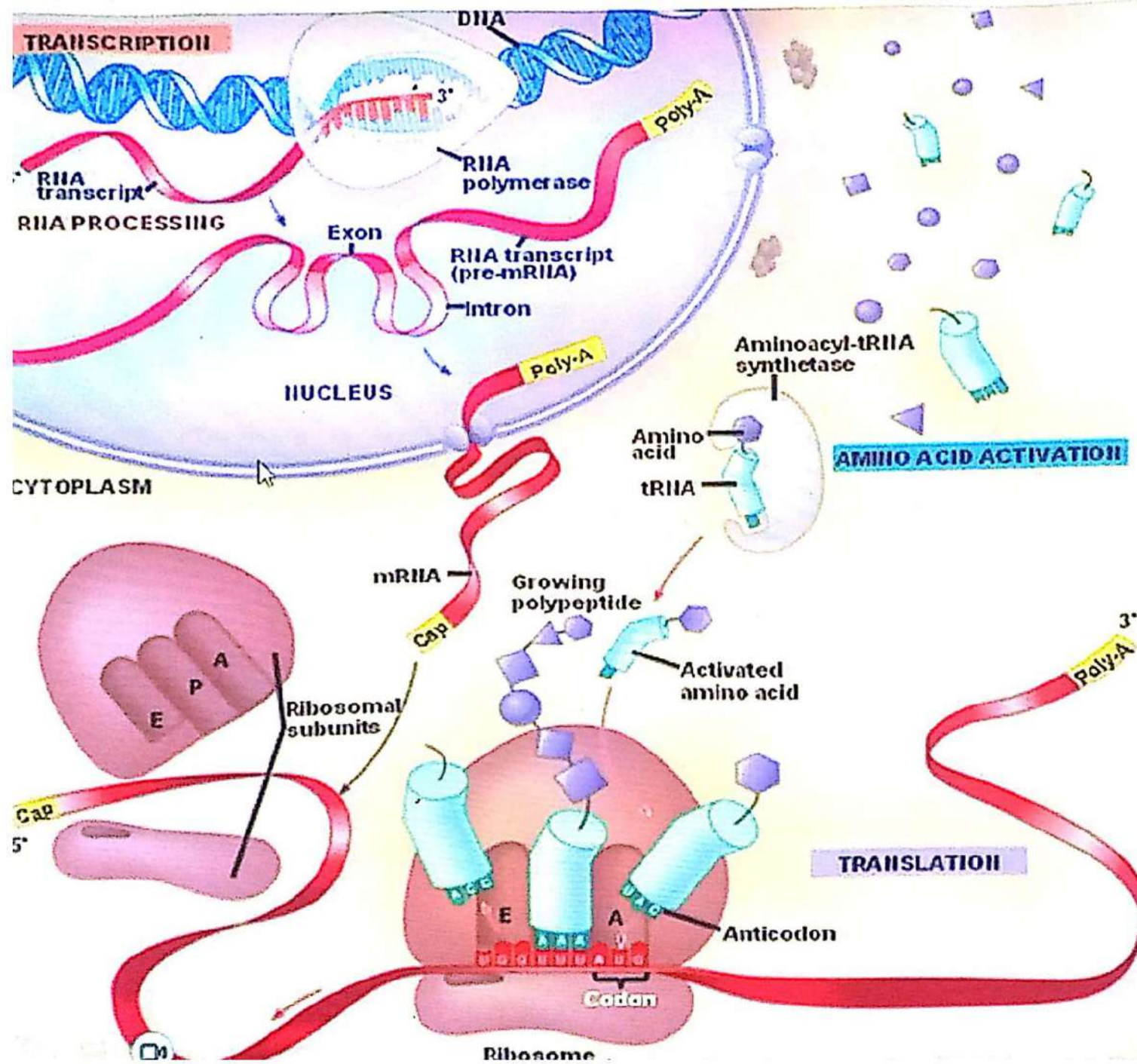


Small ribosomal subunit



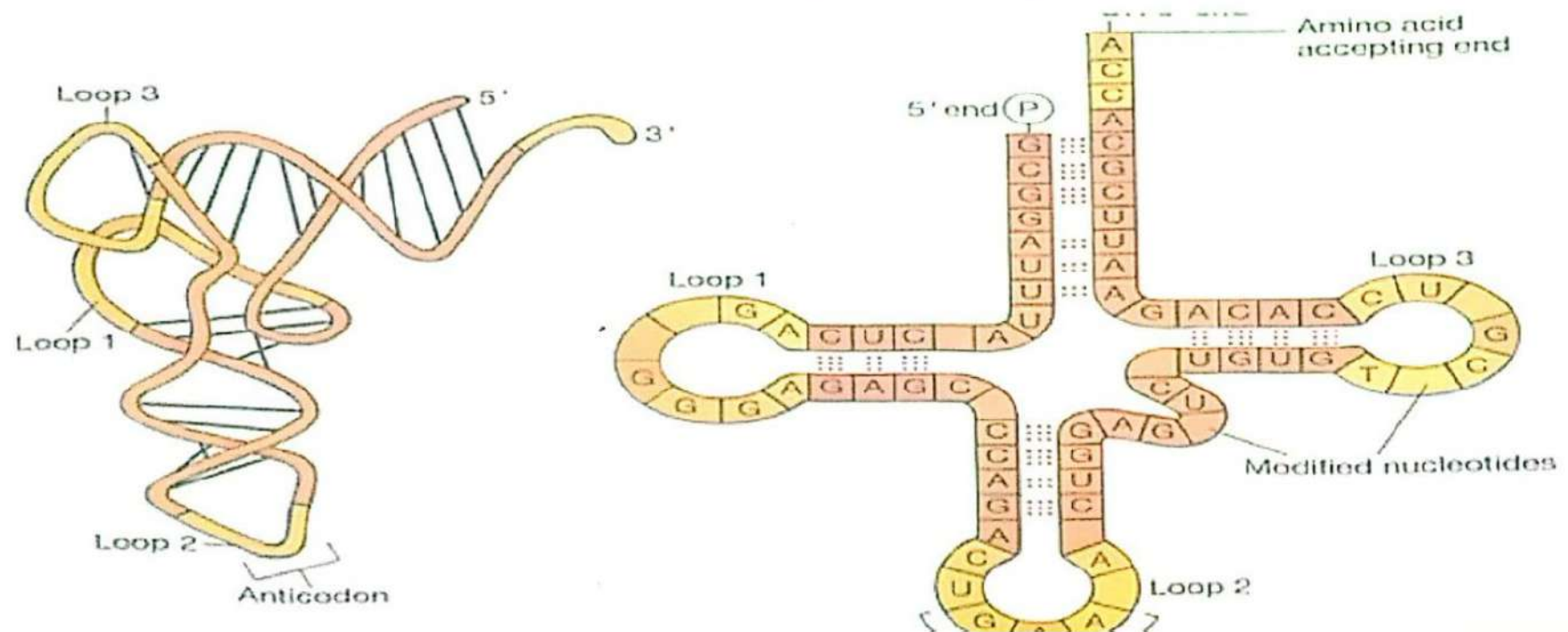
Messenger RNA (mRNA)

- ❖ Messenger RNA molecules carry chemical information from the gene of the DNA to the ribosome, where translation into protein occurs.
- ❖ They vary considerably in length and function according to the size of the gene serving as template for transcription of mRNA
- ❖ And the size of, or the number of amino acids in the protein coded by a certain gene.



Transfer RNA (tRNA)

The **smallest class** of RNA molecules, **carries amino acids** to the ribosome during translation. More than one tRNA molecule interacts with the ribosome at the same time. This may account for the relative small size of tRNA. All tRNAs has a **two-dimensional cloverleaf model**,



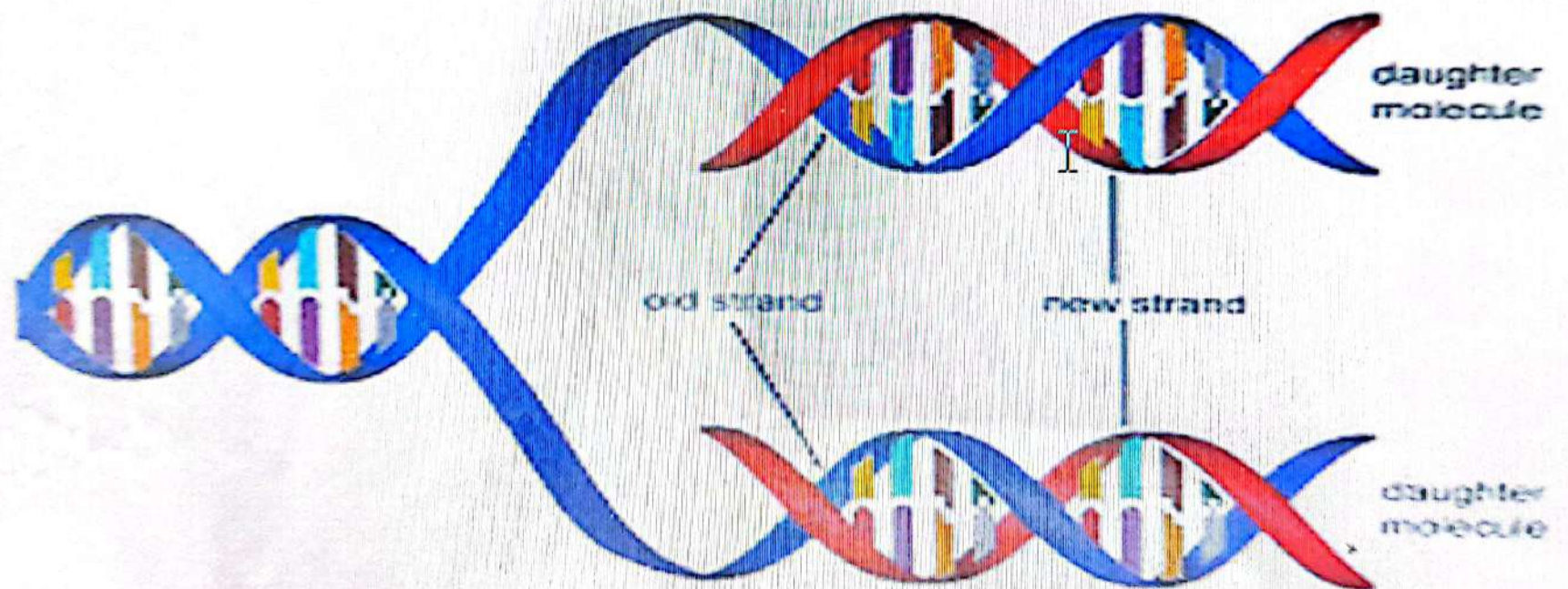
DNA REPLICATION

- Replication is an essential function of the genetic material

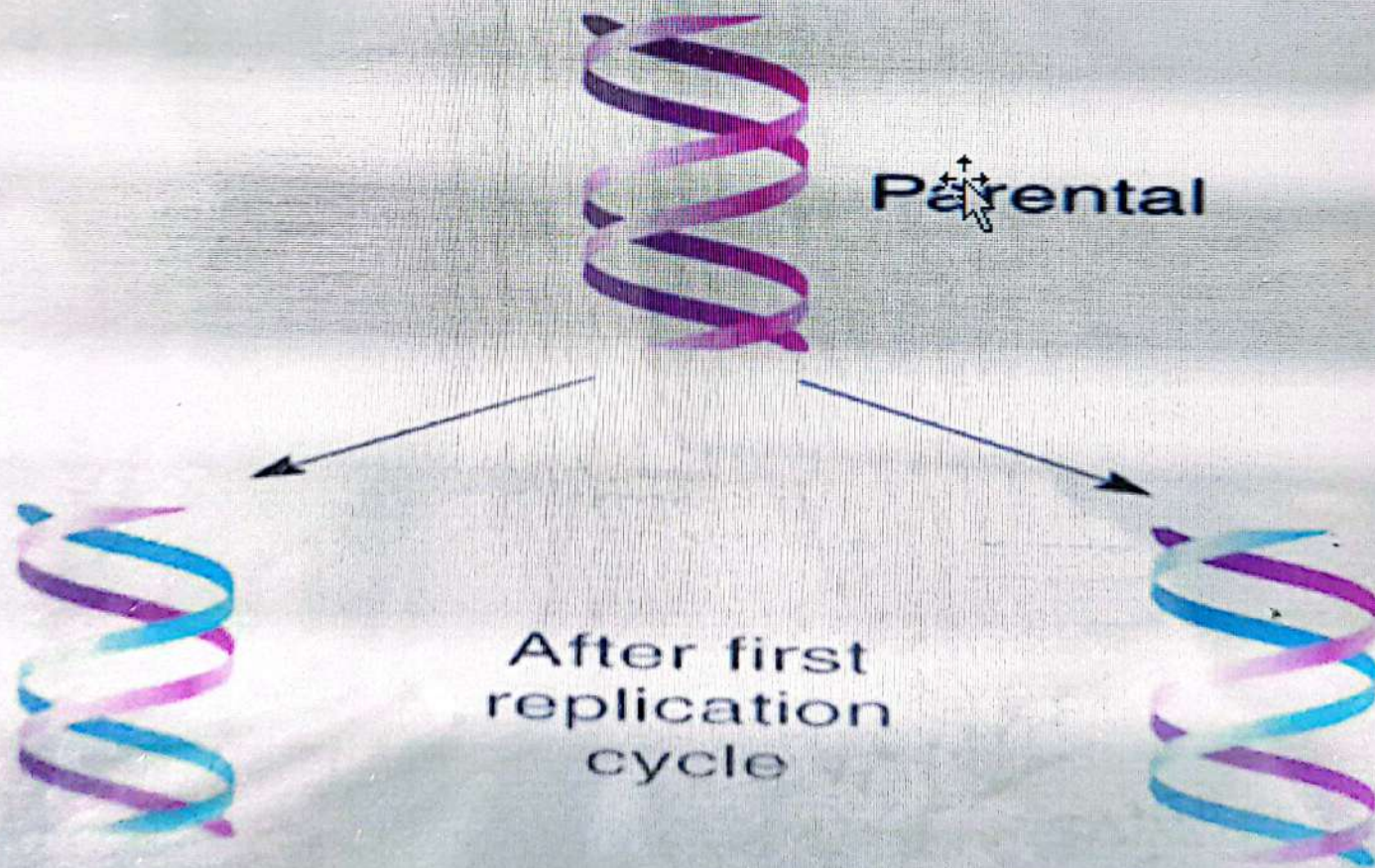
Mode of replication

1. Semi Conservative Replication

- ✓ Each replicated DNA molecule would consist of an old and a new strand.



a) Semiconservative model



DNA Synthesis

- There are three Enzymes responsible for DNA synthesis
- They synthesizes DNA in a 5' to 3' mode

1. DNA Polymerase I

- Isolated from E.coli
- Can synthesis DNA in a cell-free, *in vitro* system
- **Requirement of DNA synthesis:**
 1. All four types of deoxyribonucleosides triphosphates (**dATP, dCTP, dGTP, dTTP**)
 2. Mg^{++} (magnesium) ions
 3. Template DNA

DNA Synthesis

- **DNA polymerase I** serves only a repair function in vivo (in the living cell).
- It can synthesis short pieces of DNA.
- This enzyme has a slow rate of synthesis, 650 nucleotides per minute, which is about 100 times less than in vivo rate.

2. DNA Polymerase II

- Isolated from E.coli,
- It synthesizes DNA in a 5' to 3' mode
- Repair gaps of less than 100 nucleotides

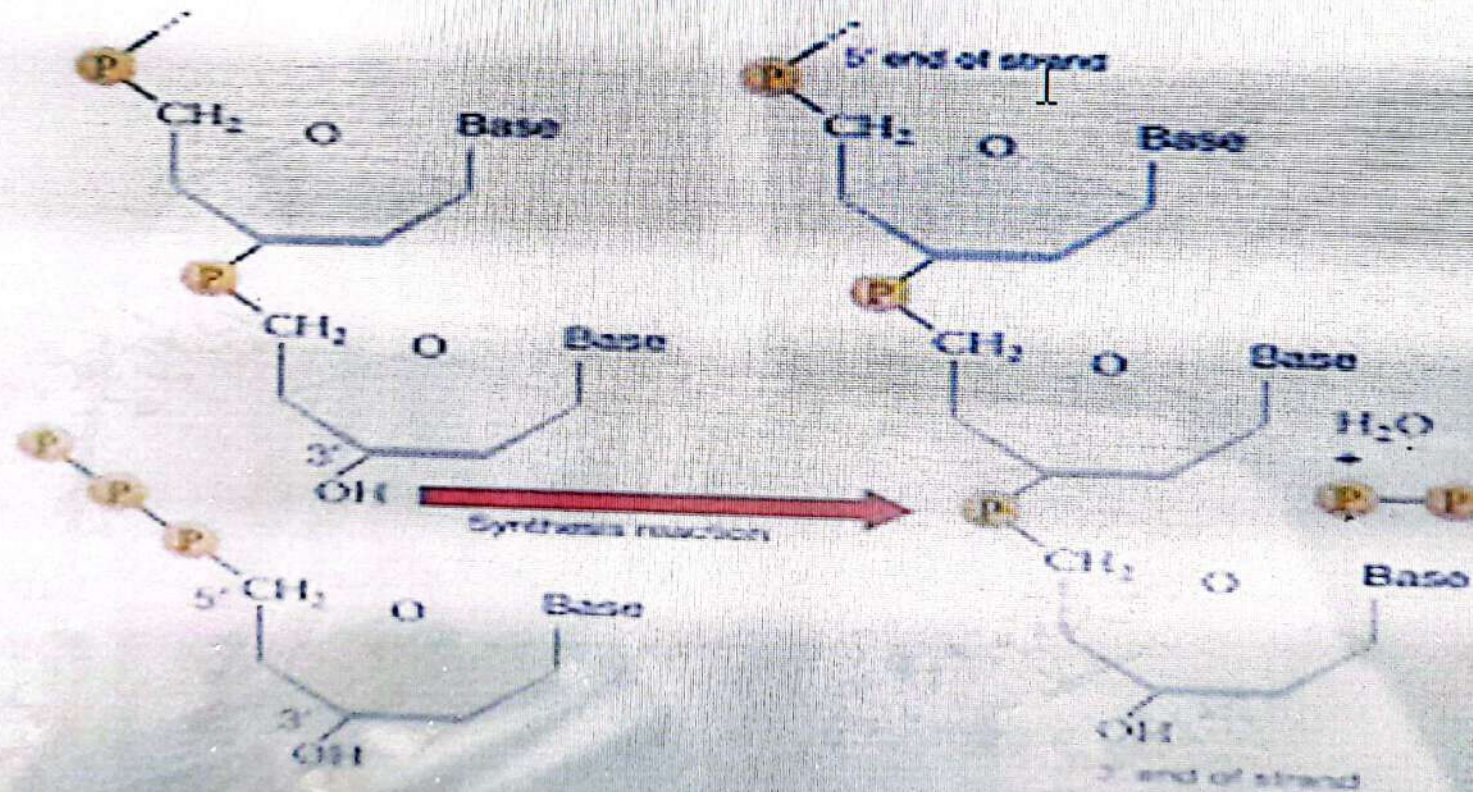
DNA Synthesis

3. DNA Polymerase III

- It requires a free 3' OH end of a polynucleotide in order to add a nucleotide.
- It is the major enzyme responsible for cellular DNA replication in *E. coli*.
- In eukaryotes, the equivalent enzyme is called DNA polymerase alpha.
- DNA Polymerase III is ATP-dependant and can add 700 nucleotides per second at 37 °C

DNA Synthesis

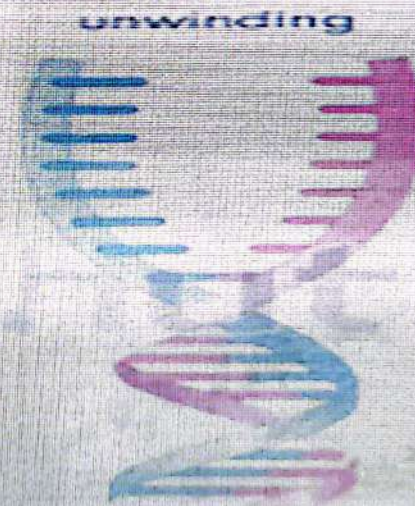
- It has an **exonuclease activity** (it can proof-read its own product, remove the mistake and then correct the error)



Steps of DNA Synthesis

1. Unwinding the DNA Helix = **Opening the double helix**

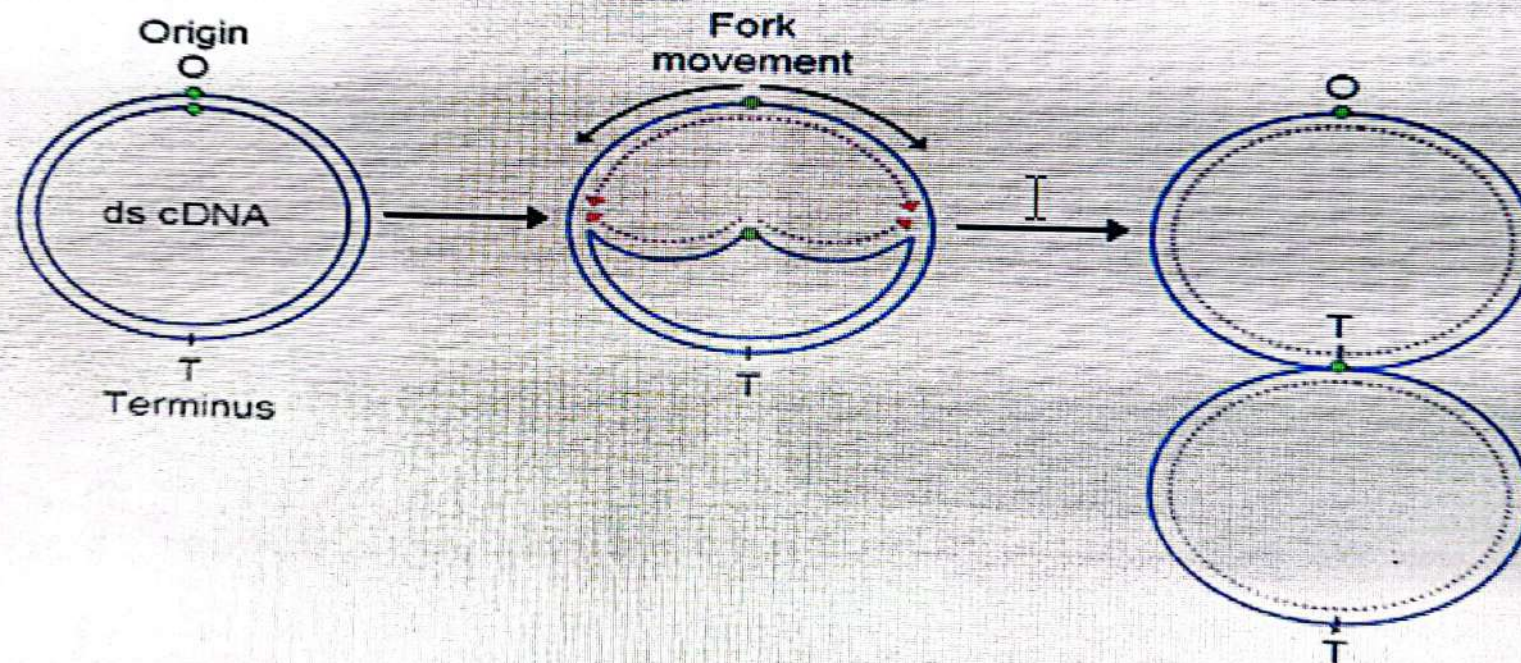
Unwinding the helix and denaturation or the separation of the two strands from each other (by breaking of the hydrogen bonds between the two complementary strands).



Unwinding the DNA Helix

- Unwinding starts at a *point of origin* following by the synthesis of two complementary stands that proceeds in both directions.
- *E. coli*, where the DNA is circular, replication starts from only one point of origin and continues in both directions until the whole chromosome is replicated.

Unwinding the DNA Helix

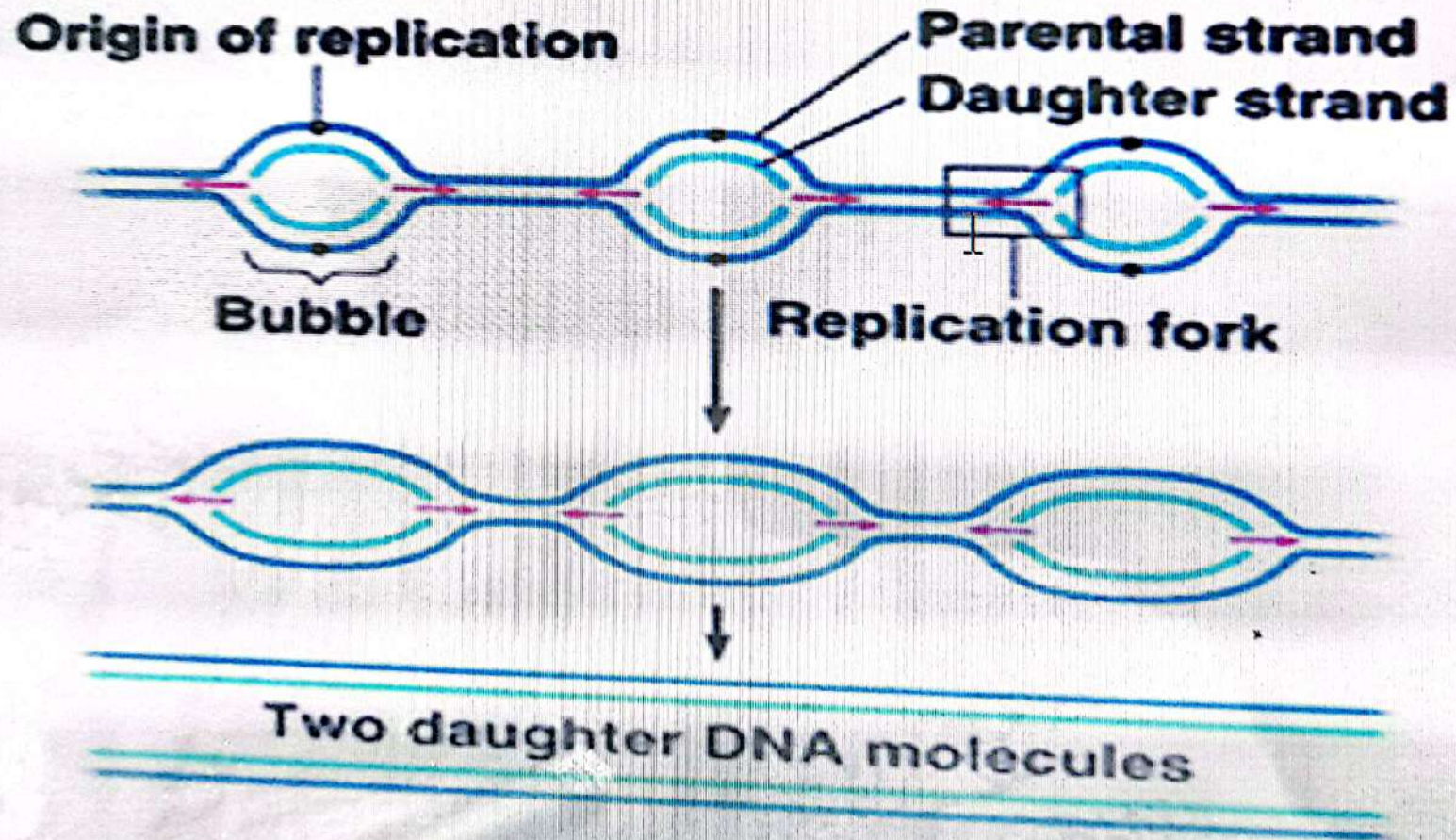


Semi-conservative replication of circular DNA in bacteria

Unwinding the DNA Helix

- o In eukaryotes, however, there are several points of origin distributed on each chromosome, and replication starts in both directions from each of these points. 
- o Each segment of the chromosome whose replication is under the control of one point of origin is called **a replicon**.
- o Thus there are many replicons per eukaryotic chromosome, each with its own origin

Unwinding the DNA Helix



Enzymes (or proteins) participate in the unwinding process

1. Helicase

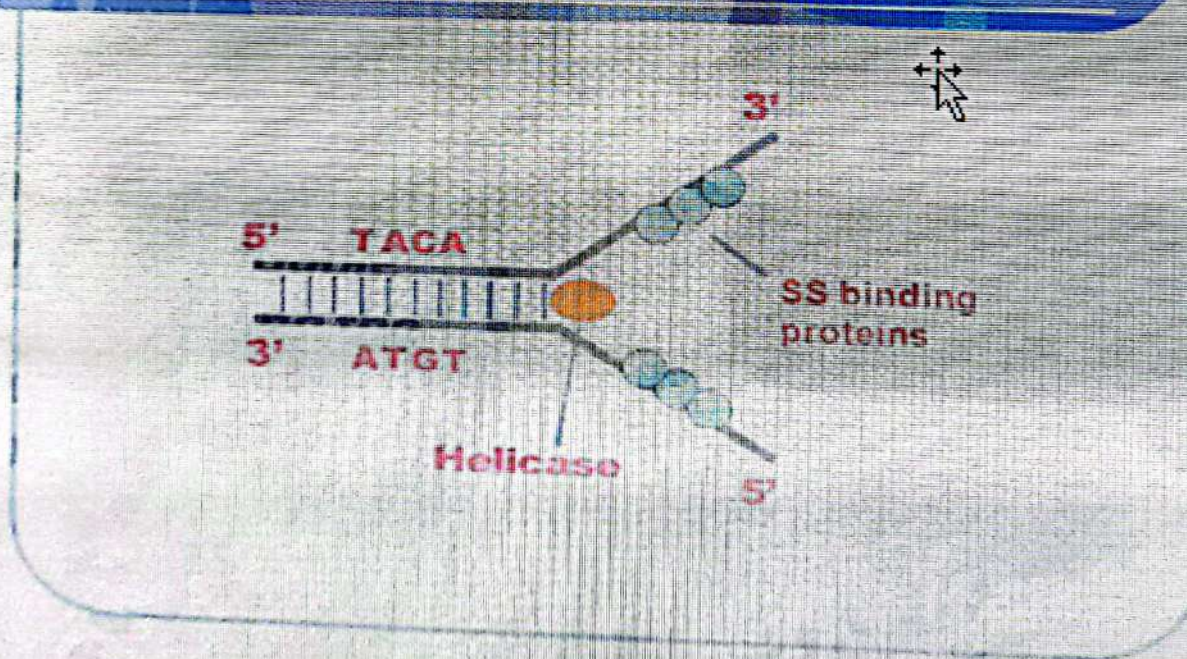
- also called unwinding proteins, initiate **localized denaturing** of the helix, causing it to unwind. This needs ATP energy

2. Single Strand Binding (SSB)

- Proteins, originally called helix destabilizing proteins (HDP): These **binds to the strands** that are created, stabilizing the template which is to be replicated.

Unwinding the DNA Helix

Step 1: Unwinding Double Helix

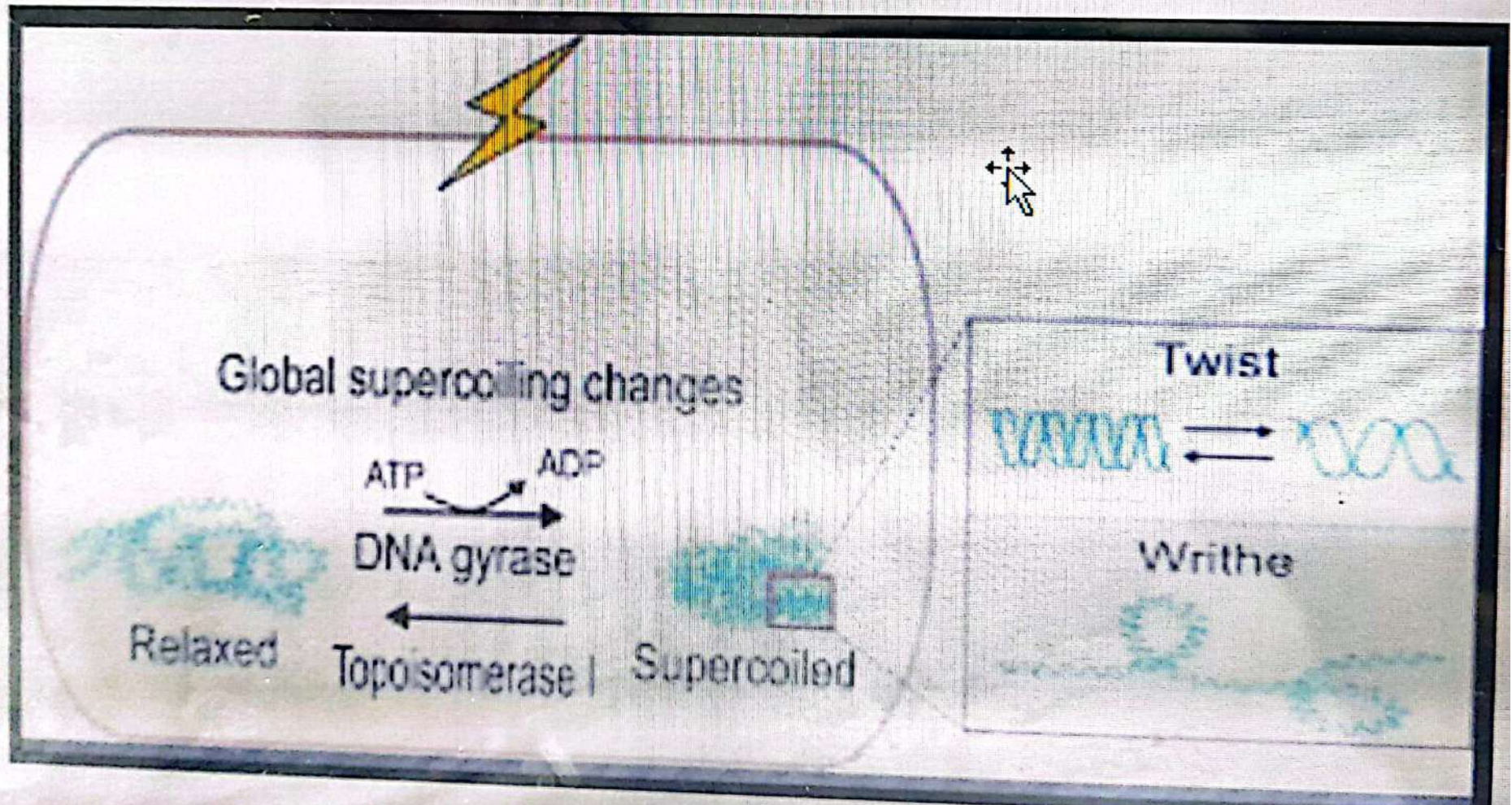


Enzymes (or proteins) participate in the unwinding process

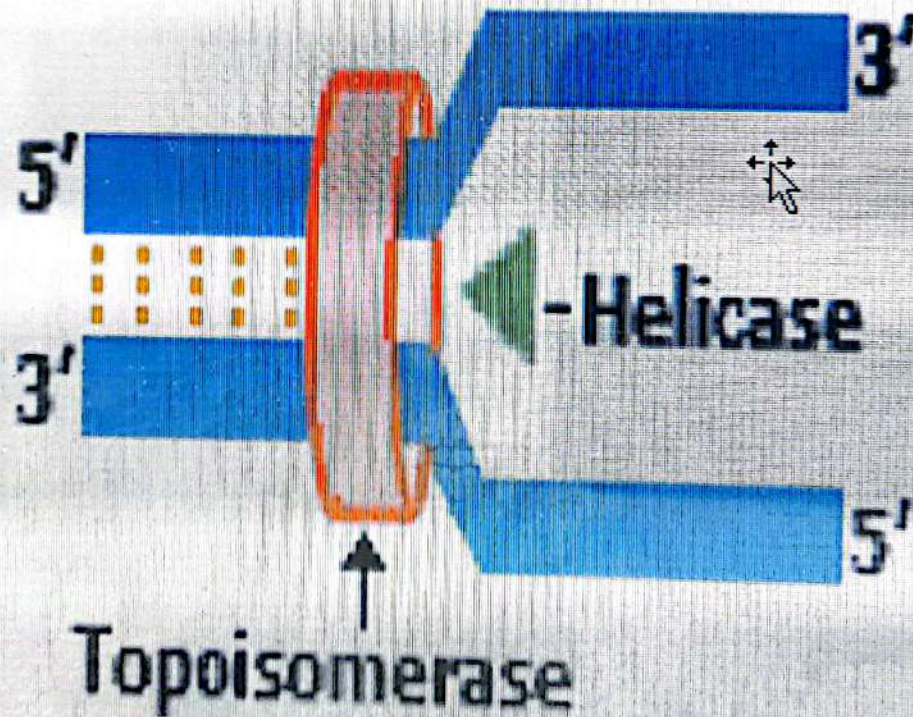
3. DNA Gyrase or Topoisomerase I

- Functions to **reduce the tension** caused by super coiling as a result of the unwinding process. 
- Release of the tension by DNA gyrase is accomplished by causing a **double-stranded break** or nicking.
- When the tension is released, the **enzyme seals or closes the**

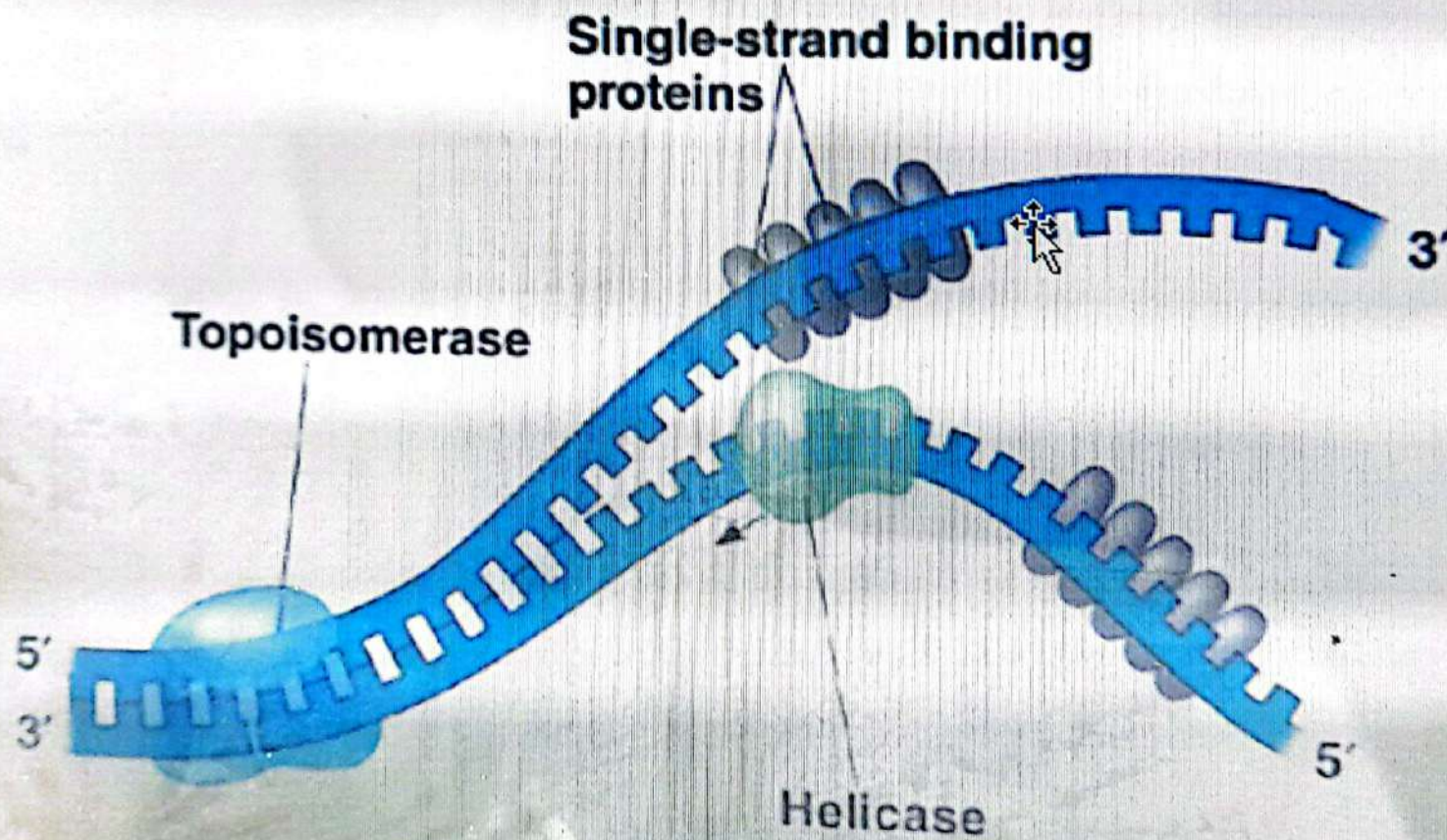
Enzymes (or proteins) participate in the unwinding process



Unwinding the DNA Helix



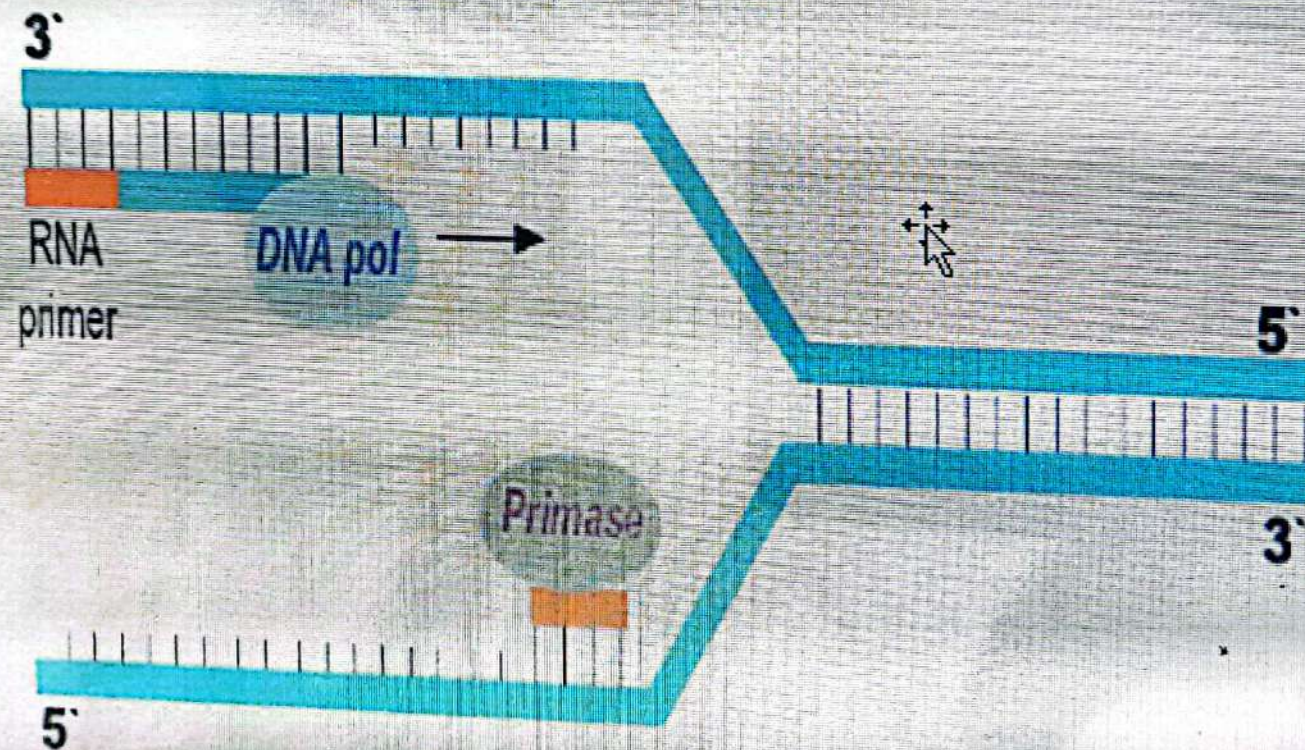
Unwinding the DNA Helix



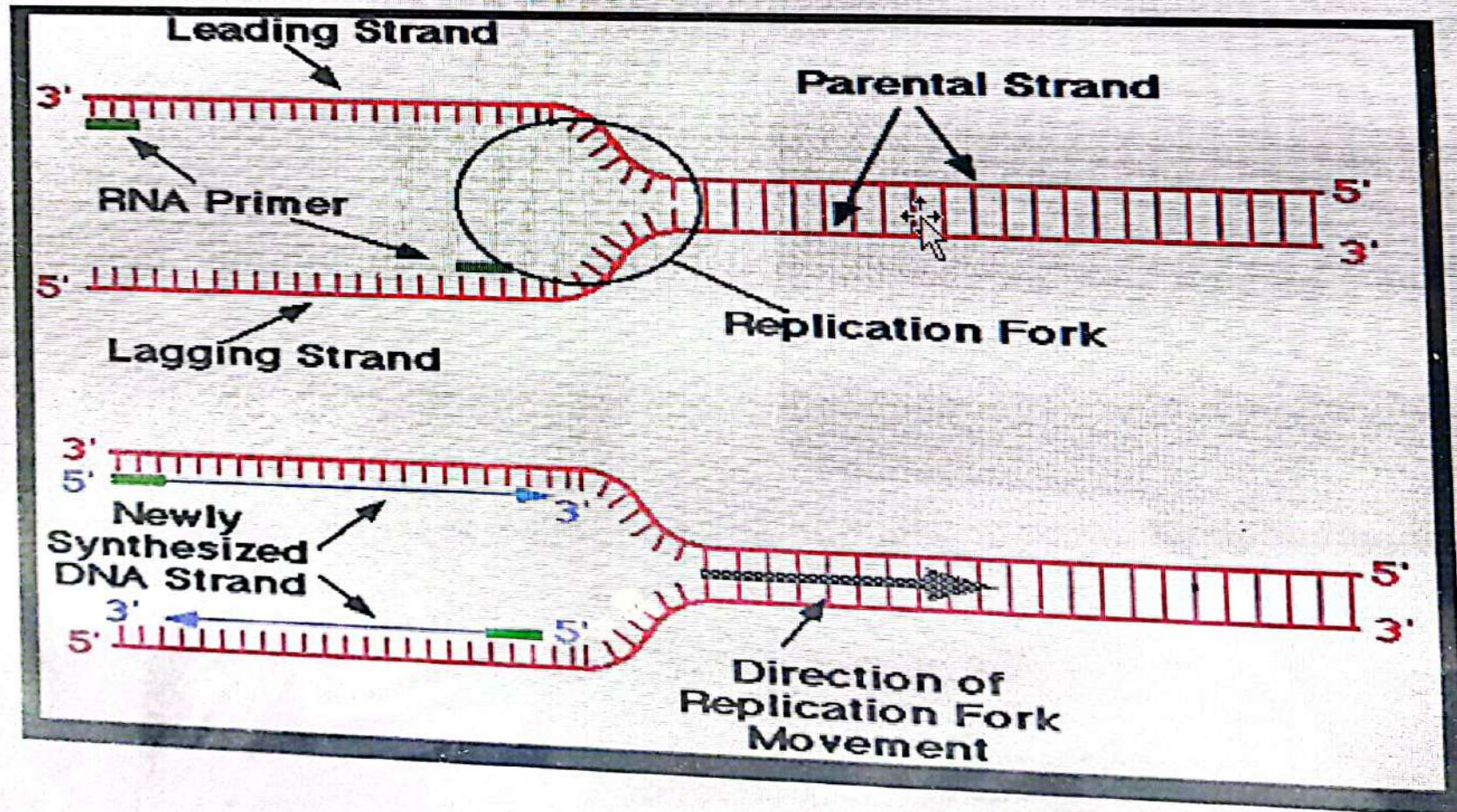
II. Initiation of DNA Synthesis

- Initiation of synthesis may occur, once a small portion of the helix has been **unwound**.
- The enzyme responsible for replication is **DNA polymerase III**.
- This enzyme requires a free **3' OH** which is provided by the synthesis of a short segment of RNA under the direction of **RNA polymerase** called **a primer**.

II. Initiation of DNA Synthesis



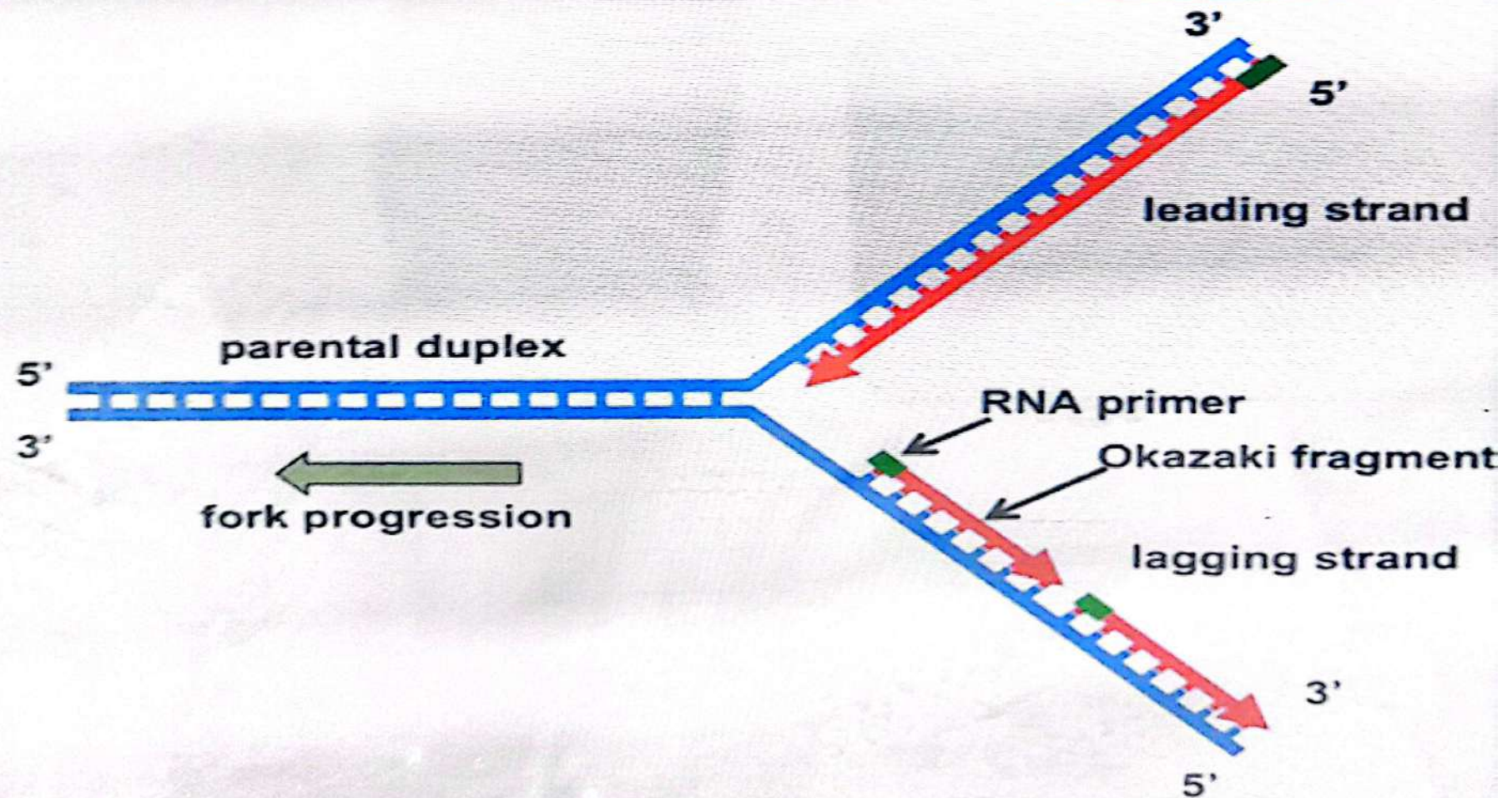
II. Initiation of DNA Synthesis



III. Continuous, Discontinuous and Bidirectional Synthesis

- The two strands of DNA double helix are **antiparallel** (one runs in a 5__-to-3__ direction, while the other has the opposite polarity).
- This structural characteristic is responsible for two features of DNA synthesis.

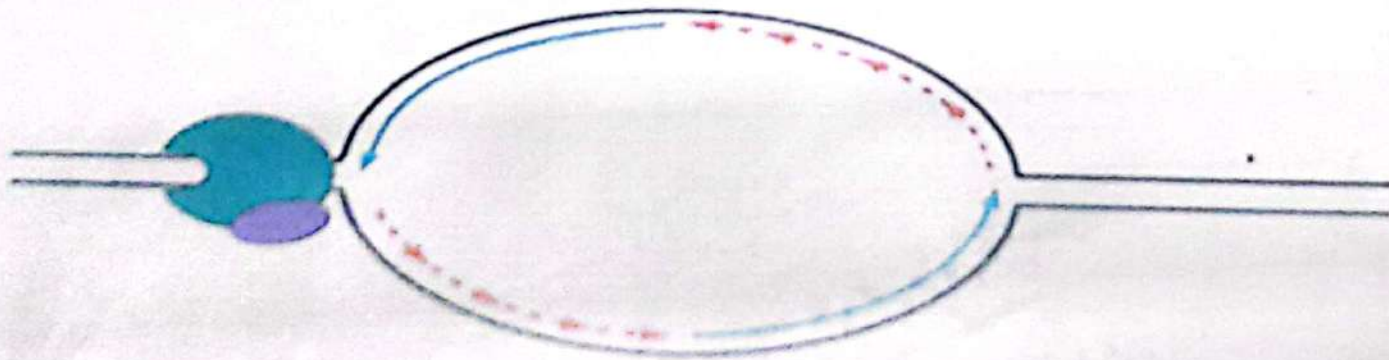
Continuous, Discontinuous DNA Synthesis



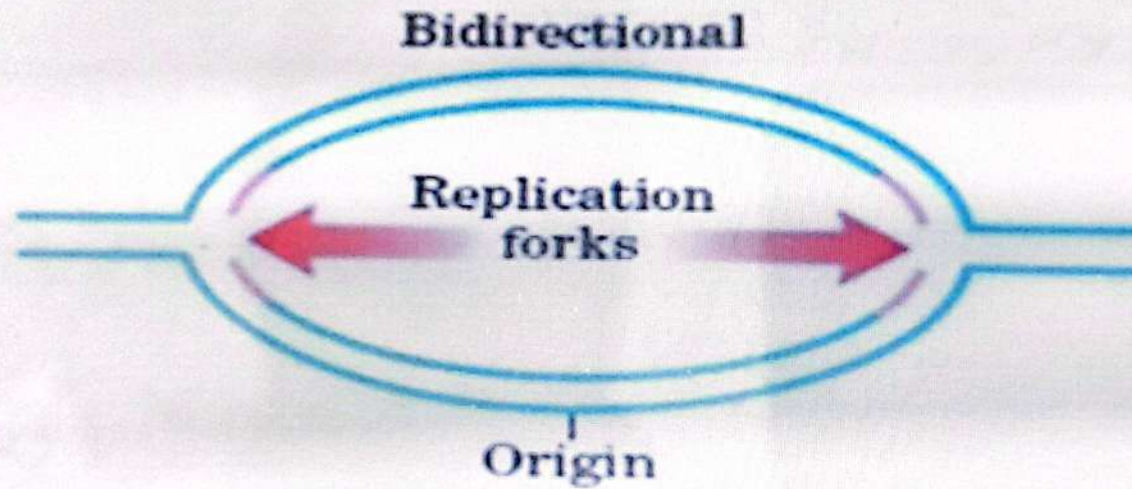
Bidirectional DNA Synthesis

- First, since DNA polymerase III synthesizes DNA in a 5' to 3' direction only, simultaneous synthesis of antiparallel strands must be bidirectional.
- That is the enzyme moves in one direction along one strand and in the opposite direction on the other

Bidirectional Replication



Bidirectional replication



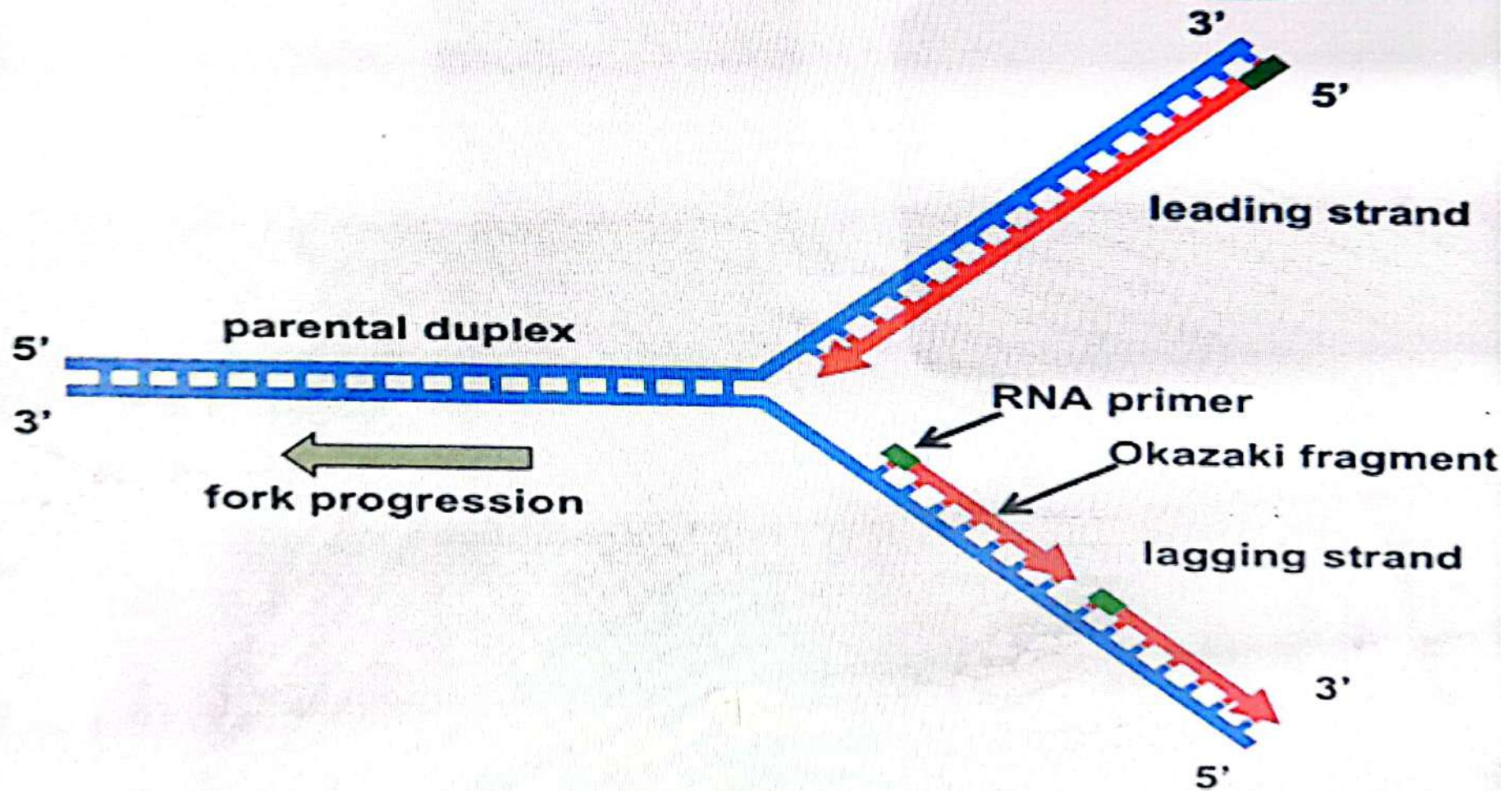
Bidirectional DNA Synthesis

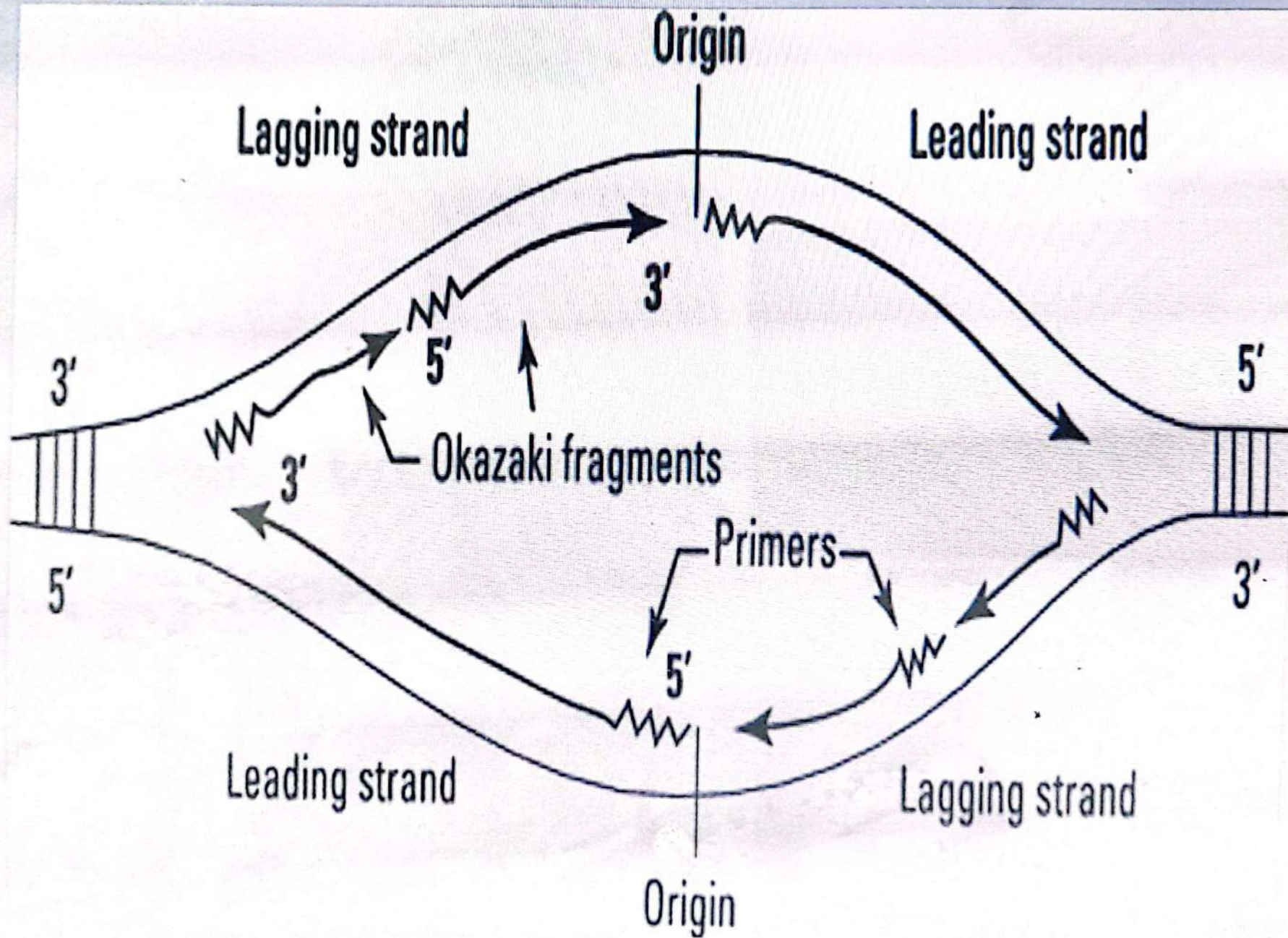


Continuous, Discontinuous DNA Synthesis

- **Second**, at the point of origin on each strand replication can be *continuous* on one side of the point of origin (on the **leading strand**) , and *discontinuous* on the other side (on the **lagging strand**).
- As the *replication fork* on each side opens up and progresses along the helix, new points of priming are exposed on the discontinuous strand.

Continuous, Discontinuous DNA Synthesis

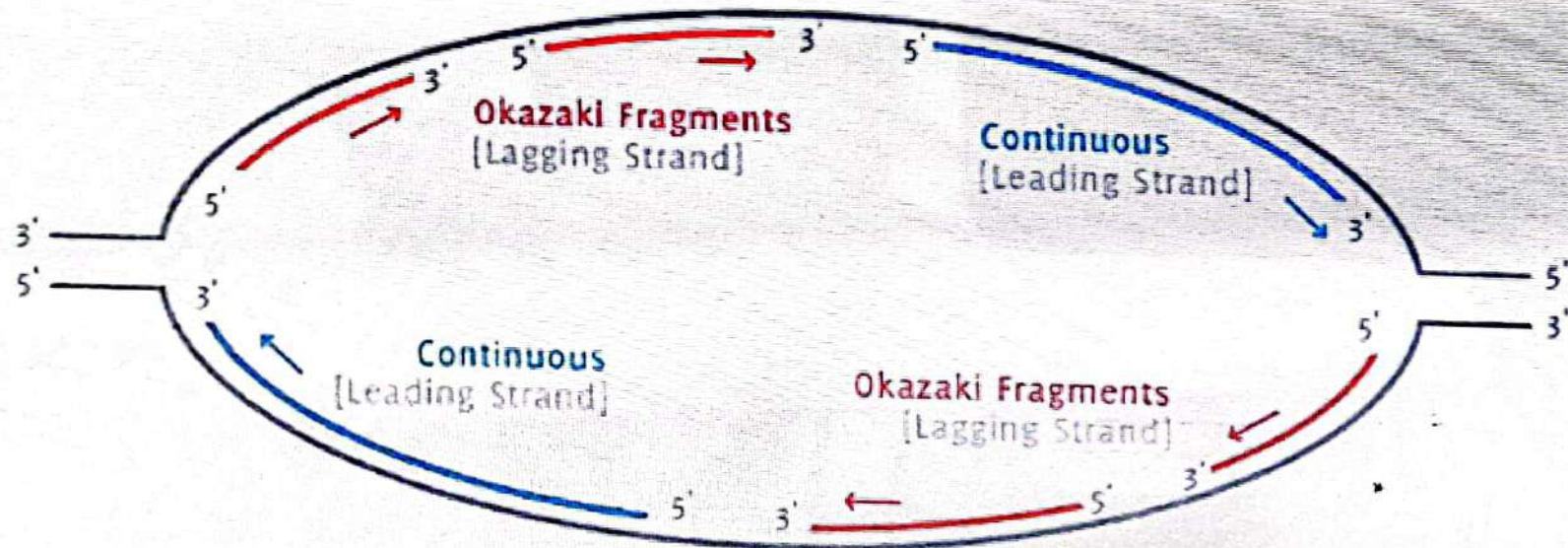


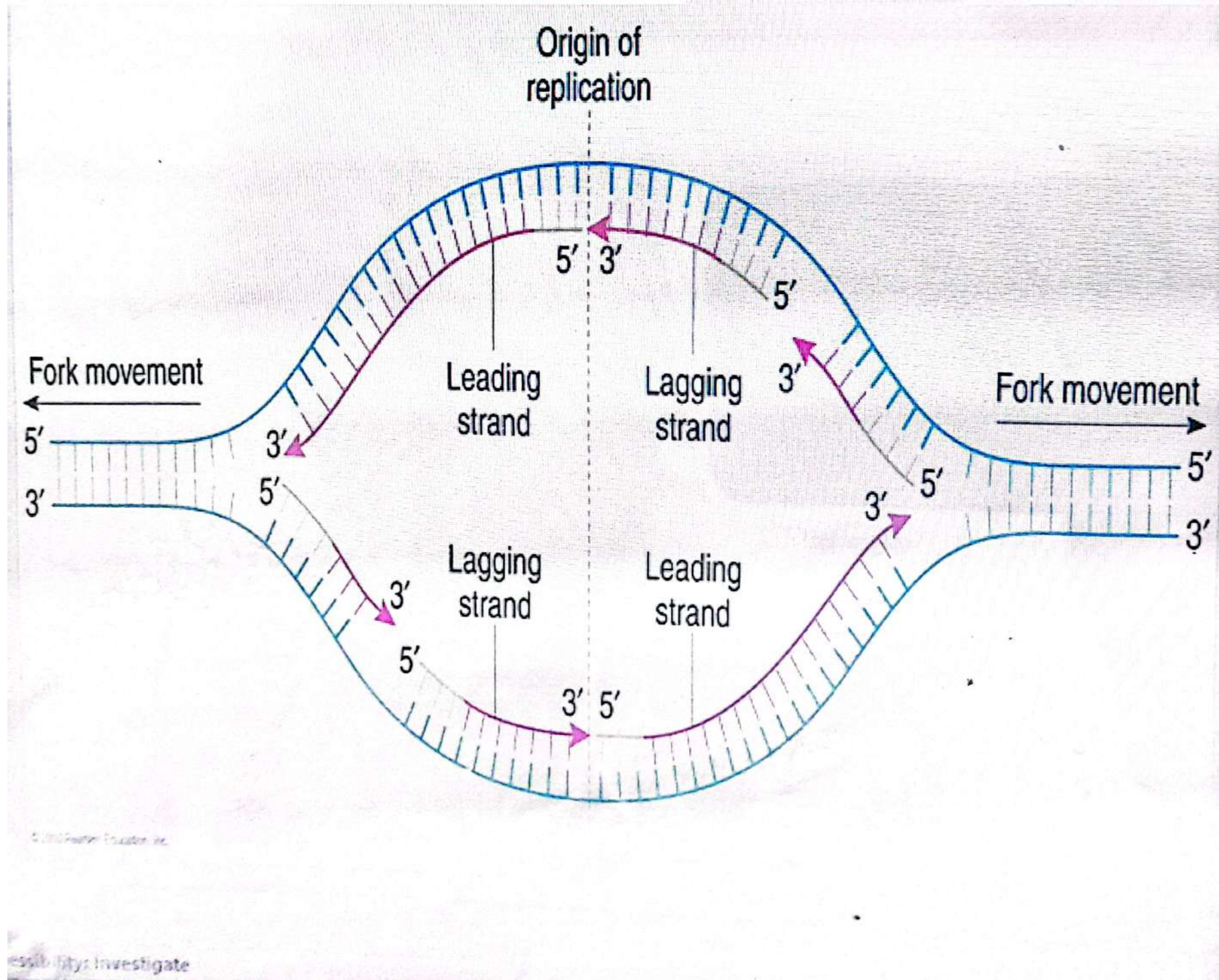


Continuous, Discontinuous DNA Synthesis

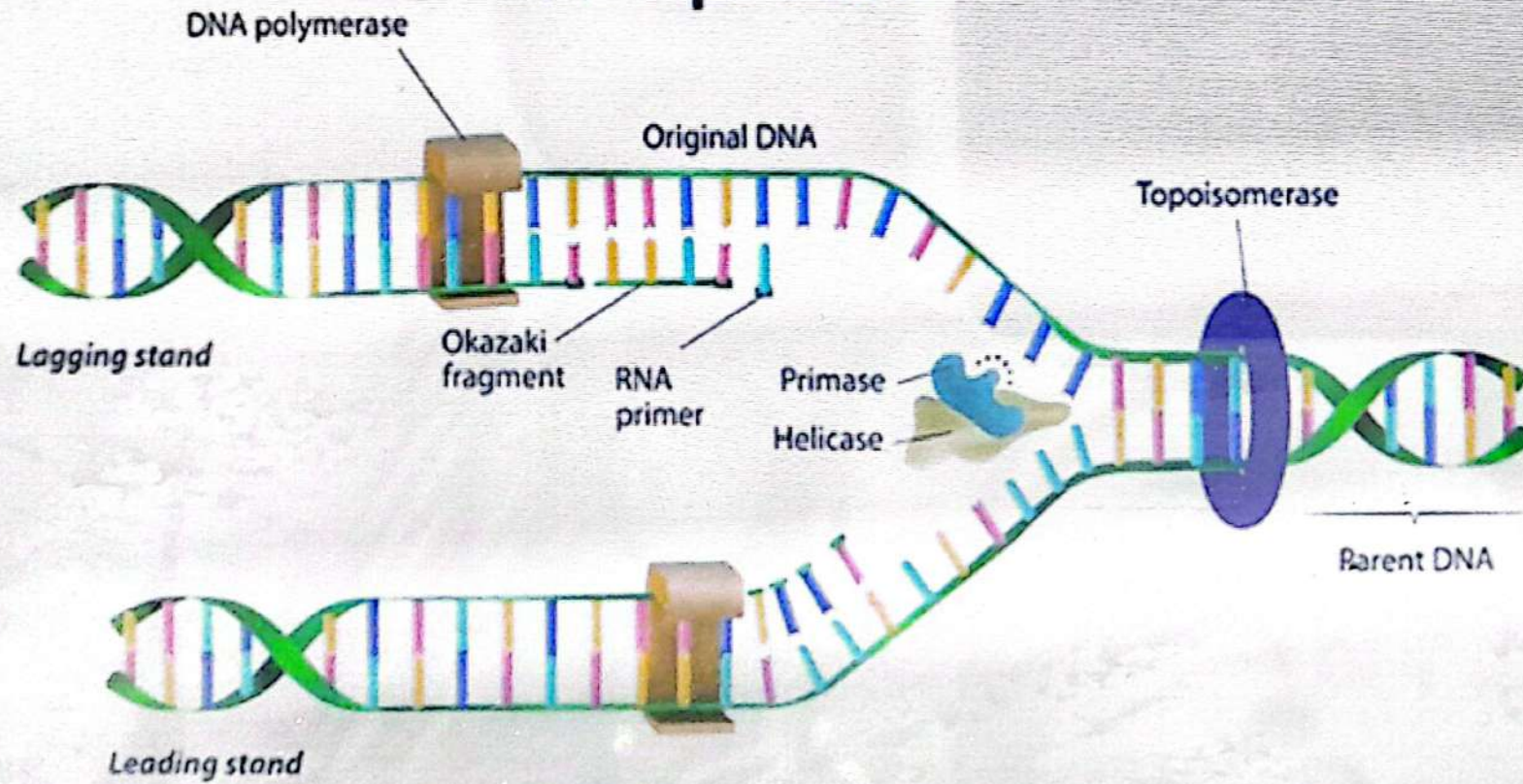
- This leads to interrupted DNA synthesis on one side of the point of origin on each strand .
- These relatively short (1000-2000 nucleotides in *E. coli* and 100-150 nucleotides in eukaryotes) are called **Okazaki fragments**
- Discontinuous synthesis requires the enzyme *DNA ligase* which units the DNA (Okazaki) fragments in a continuous strand.
- Synthesis ends when the newly synthesized strand joins with that of a neighbouring replicon.

Continuous, Discontinuous DNA Synthesis





DNA replication



Model of DNA Synthesis

4. **DNA polymerase III** (polymerase alpha in eukaryotes) synthesizes the complementary strands of DNA in the 5'-to-3' direction.

5. **Synthesis is bidirectional.**

6. On one side of the point of origin in each strand, synthesis is *discontinuous* and produces **Okazaki fragments**.

7. Okazaki fragments are sealed by **DNA ligase**.

8. **Primer DNA** is removed by DNA polymerase I, which synthesizes short DNA in place of primer..

Model of DNA Synthesis

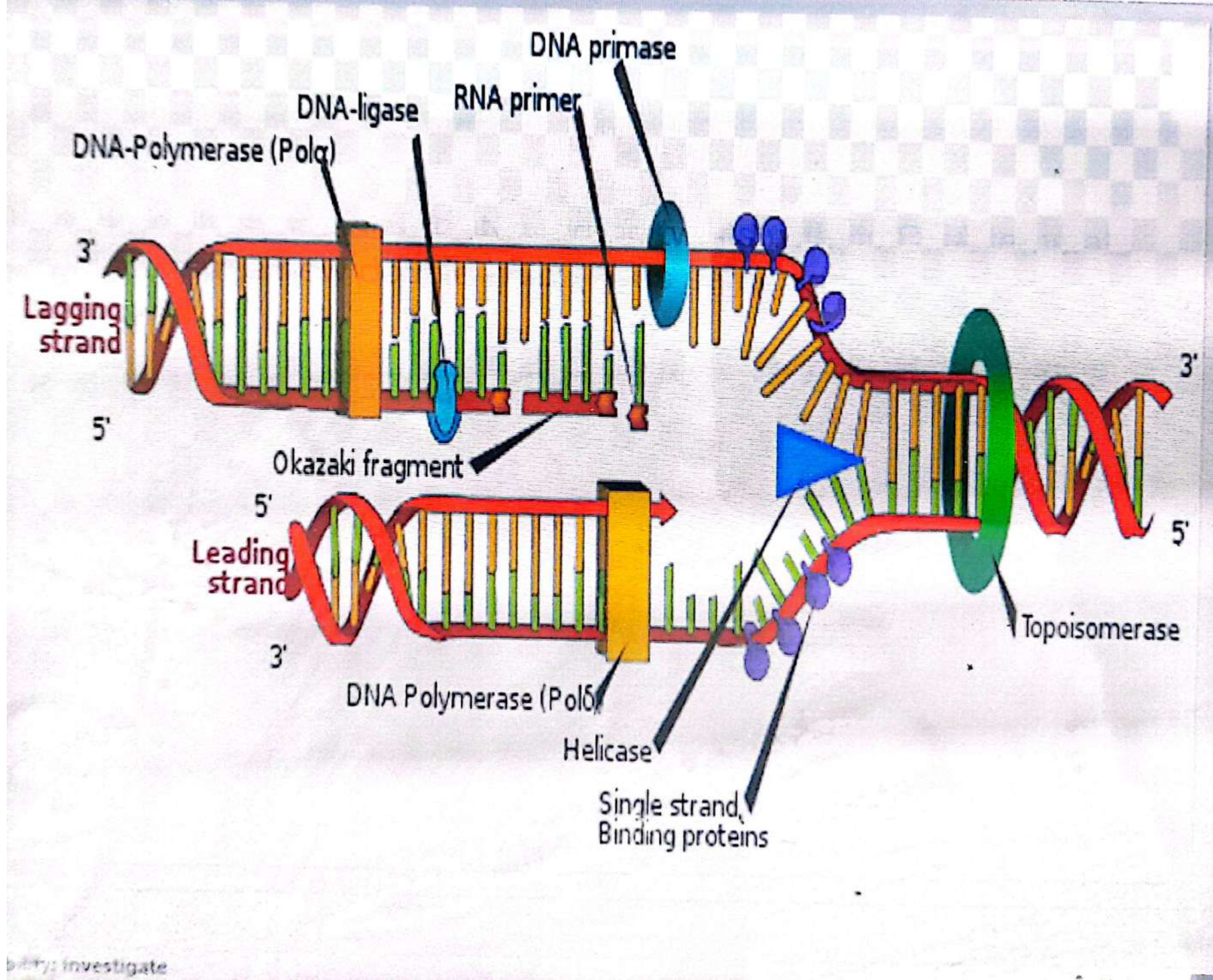
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Types of DNA Synthesis

1. Unique or Single Copy Sequences

- 57% of total DNA in cattle
- Dispersed throughout the genome
- Average length of around 1000 bases.
- A small proportion of a single copy DNA account for most genes

2. Interspersed-repetitive DNA

- A round 300 bases
- Occurring up to 10^4 to 10^5 times in the genome.
- 20% of bovine DNA.
- Dispersed throughout the genome

Types of DNA Synthesis

o 3. **Satellite DNA**

- o Satellite DNA is highly repetitive
- o one or a small number of short repeat unit
- o occurring a very large number of times (10^6 or more) .
- o repeated tandemly, not dispersed throughout the genome..
- o It is located in regions of constitutive heterochromatin.
- o Bovine DNA contains eight different satellites which together constitute around 23% of total bovine DNA.



**Thank
you!**